

# LECTURE NOTES ON INTEGRABLE SYSTEMS

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ABSTRACT. These notes are based on the course *Introduction to Integrable Systems*, delivered by Professor Liming Ling during the Fall semester of 2025–2026.

The full L<sup>A</sup>T<sub>E</sub>X source code of these notes is available at: <https://codeberg.org/killlkill/courseNotes>.

## Week 1: Algebro-Geometric Solutions for the mKdV Equation (1)

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**Objectives:** Derive the mKdV equation from the associated spectral problem and construct the AKNS hierarchy by explicit recursion.

The focusing modified Korteweg–de Vries (mKdV) equation is given by

$$(1.1) \quad u_t + 6u^2u_x + u_{xxx} = 0, \quad u(x, t) \in \mathbb{R}, \quad (x, t) \in \mathbb{R}^2.$$

A standard method for establishing integrability is via a Lax pair. We introduce a matrix function  $\Phi(x, t)$  satisfying

$$\Phi_x = U\Phi, \quad \Phi_t = V\Phi,$$

where  $U$  and  $V$  are  $2 \times 2$  matrices depending on  $u$  and its derivatives. The compatibility condition

$$\Phi_{xt} = \Phi_{tx} \iff U_t - V_x + [U, V] = 0,$$

with  $[U, V] = UV - VU$ , is equivalent to the zero-curvature equation.

Following the AKNS formalism, the spatial part of the Lax pair is taken as

$$(1.2) \quad \Phi_x = U\Phi, \quad U = -i\lambda\sigma_3 + Q,$$

where

$$(1.3) \quad Q = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \quad \sigma_3 = \text{diag}(1, -1).$$

We introduce the transformation

$$(1.4) \quad \Phi(x; \lambda) = m(x; \lambda)e^{-i\lambda x\sigma_3}, \quad m(x; \lambda) \rightarrow \mathbb{I} \text{ as } \lambda \rightarrow \infty.$$

Substituting into the spectral problem (1.2) gives

$$(1.5) \quad \Phi_x \Phi^{-1} = m_x m^{-1} - i\lambda m \sigma_3 m^{-1} = -i\lambda \sigma_3 + Q.$$

Define

$$(1.6) \quad \Psi = m \sigma_3 m^{-1}.$$

Differentiating  $\Psi$  and using (1.5), we compute

$$\begin{aligned} \Psi_x &= m_x \sigma_3 m^{-1} - m \sigma_3 m^{-1} m_x m^{-1} \\ &= [m_x m^{-1}, m \sigma_3 m^{-1}] \\ &= [m_x m^{-1} - i\lambda m \sigma_3 m^{-1}, m \sigma_3 m^{-1}] \\ &= [U, \Psi]. \end{aligned}$$

Thus  $\Psi$  satisfies

$$(1.7) \quad \Psi_x = [U, \Psi],$$

which is the stationary zero-curvature equation.

The matrix  $m(x; \lambda)$  admits the asymptotic expansion

$$(1.8) \quad m(x; \lambda) = \mathbb{I} + \lambda^{-1}m_1 + \lambda^{-2}m_2 + \dots.$$

From (1.6) and (1.8) we obtain the formal expansion

$$(1.9) \quad \Psi = \Psi_0 + \Psi_1\lambda^{-1} + \Psi_2\lambda^{-2} + \dots, \quad \Psi_0 = \sigma_3.$$

Substituting this into (1.7) yields

$$\begin{aligned} \sum_{j=1}^{\infty} \Psi_{j,x} \lambda^{-j} &= -i\lambda \left[ \sigma_3, \sum_{j=1}^{\infty} \Psi_j \lambda^{-j} \right] + \left[ Q, \sum_{j=0}^{\infty} \Psi_j \lambda^{-j} \right] \\ &= -i \sum_{j=0}^{\infty} \left[ \sigma_3, \Psi_{j+1} \right] \lambda^{-j} + \sum_{j=0}^{\infty} \left[ Q, \Psi_j \right] \lambda^{-j}. \end{aligned}$$

At order  $O(1)$  we find

$$-i[\sigma_3, \Psi_1] + [Q, \sigma_3] = 0 \iff [i\Psi_1 + Q, \sigma_3] = 0,$$

which implies

$$\Psi_1^{\text{off}} = iQ.$$

For general  $j \geq 1$ , the coefficient of  $\lambda^{-j}$  gives

$$(1.10) \quad \Psi_{j,x} = -i[\sigma_3, \Psi_{j+1}] + [Q, \Psi_j] = -2i\sigma_3\Psi_{j+1}^{\text{off}} + [Q, \Psi_j].$$

Hence

$$(1.11) \quad \Psi_{j,x}^{\text{off}} = -2i\sigma_3\Psi_{j+1}^{\text{off}} + [Q, \Psi_j^{\text{diag}}] \iff \Psi_{j+1}^{\text{off}} = \frac{i}{2}\sigma_3(\Psi_{j,x}^{\text{off}} - [Q, \Psi_j^{\text{diag}}]),$$

and

$$(1.12) \quad \Psi_{j,x}^{\text{diag}} = [Q, \Psi_j^{\text{off}}] \iff \Psi_j^{\text{diag}} = \partial_x^{-1}([Q, \Psi_j^{\text{off}}]).$$

Thus, the recursion relation (1.11) provides a direct way to compute the off-diagonal coefficients algebraically in terms of lower-order terms, while the diagonal recursion (1.12) requires performing an integration. In practice, this makes the off-diagonal part more convenient to obtain explicitly, whereas the diagonal part is generally harder to compute in closed form.

By (1.6), the additional constraint  $\Psi^2 = \mathbb{I}$  provides algebraic relations among the coefficients. Substituting the expansion (1.9) yields

$$\mathbb{I} = \left( \sum_{j=0}^{\infty} \Psi_j \lambda^{-j} \right)^2 = \sum_{j=0}^{\infty} \left( \sum_{k=0}^j \Psi_k \Psi_{j-k} \right) \lambda^{-j},$$

so that

$$(1.13) \quad \sum_{k=0}^j \Psi_k \Psi_{j-k} = 0, \quad j \geq 1.$$

Separating diagonal and off-diagonal parts, for  $j \geq 1$  we obtain

$$(1.14) \quad \sum_{k=0}^j (\Psi_k^{\text{diag}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k}^{\text{off}}) = 0,$$

$$(1.15) \quad \sum_{k=0}^j (\Psi_k^{\text{off}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{diag}} \Psi_{j-k}^{\text{off}}) = 0.$$

Since  $\Psi_0 = \sigma_3$ , the off-diagonal constraint (1.15) for  $j \geq 2$  takes the form

$$\sigma_3 \Psi_j^{\text{off}} + \Psi_j^{\text{off}} \sigma_3 + \sum_{k=1}^{j-1} (\Psi_k^{\text{off}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{diag}} \Psi_{j-k}^{\text{off}}) = 0.$$

Noting that  $\sigma_3 \Psi_j^{\text{off}} + \Psi_j^{\text{off}} \sigma_3 = 0$ , this simplifies to

$$(1.16) \quad \sum_{k=1}^{j-1} \left( \Psi_k^{\text{off}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{diag}} \Psi_{j-k}^{\text{off}} \right) = 0.$$

It should be noted that from the off-diagonal constraint (1.15) one can only deduce the reduced form (1.16), which merely expresses a compatibility condition between lower-order coefficients. In particular, this relation does not provide a practical means to compute the diagonal parts explicitly.

For the diagonal part (1.14) we have

$$\sigma_3 \Psi_j^{\text{diag}} + \Psi_j^{\text{diag}} \sigma_3 + \sum_{k=1}^{j-1} \left( \Psi_k^{\text{diag}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k}^{\text{off}} \right) = 0, \quad j \geq 2.$$

Since  $\sigma_3 \Psi_j^{\text{diag}} = \Psi_j^{\text{diag}} \sigma_3$ , this simplifies to

$$(1.17) \quad \Psi_j^{\text{diag}} = -\frac{1}{2} \sigma_3 \sum_{k=1}^{j-1} \left( \Psi_k^{\text{diag}} \Psi_{j-k}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k}^{\text{off}} \right), \quad j \geq 2.$$

We will use this reduced relation (1.17) to compute the diagonal part  $\Psi_j^{\text{diag}}$  recursively at each order. This provides an explicit algebraic formula for the diagonal coefficients in terms of lower-order diagonal and off-diagonal terms.

For  $j = 1$ , the condition (1.13) reads

$$\sigma_3 \Psi_1 + \Psi_1 \sigma_3 = 2\sigma_3 \Psi_1^{\text{diag}} = 0,$$

so that  $\Psi_1 = \Psi_1^{\text{off}} = iQ$ .

Using (1.11) and (1.17) recursively, we find

$$\Psi_2^{\text{off}} = -\frac{1}{2} \sigma_3 Q_x, \quad \Psi_2^{\text{diag}} = \frac{1}{2} \sigma_3 Q^2,$$

and hence

$$\Psi_2 = \frac{1}{2} \sigma_3 (Q^2 - Q_x).$$

For the next coefficient, we compute

$$\begin{aligned} \Psi_3^{\text{off}} &= \frac{i}{2} \sigma_3 (\Psi_{2,x}^{\text{off}} - [Q, \Psi_2^{\text{diag}}]) \\ &= \frac{i}{2} \sigma_3 \left( -\frac{1}{2} \sigma_3 Q_{xx} - [Q, \frac{1}{2} \sigma_3 Q^2] \right) \\ &= \frac{i}{2} \sigma_3 \left( -\frac{1}{2} \sigma_3 Q_{xx} - \frac{1}{2} (Q \sigma_3 Q^2 - \sigma_3 Q^2 Q) \right) \\ &= -\frac{i}{4} Q_{xx} + \frac{i}{2} Q^3. \end{aligned}$$

For the diagonal part,

$$\begin{aligned} \Psi_3^{\text{diag}} &= -\frac{1}{2} \sigma_3 \left( \Psi_1^{\text{off}} \Psi_2^{\text{off}} + \Psi_2^{\text{off}} \Psi_1^{\text{off}} + \Psi_1^{\text{diag}} \Psi_2^{\text{diag}} + \Psi_2^{\text{diag}} \Psi_1^{\text{diag}} \right) \\ &= -\frac{1}{2} \sigma_3 \left( (iQ) \left( -\frac{1}{2} \sigma_3 Q_x \right) + \left( -\frac{1}{2} \sigma_3 Q_x \right) (iQ) \right) \\ &= -\frac{1}{2} \sigma_3 \left( -\frac{i}{2} (Q \sigma_3 Q_x + \sigma_3 Q_x Q) \right) \\ &= \frac{i}{4} (Q_x Q - Q Q_x) = 0. \end{aligned}$$

Proceeding to  $\Psi_4$ , the off-diagonal recursion (1.11) with  $\Psi_3^{\text{diag}} = 0$  gives

$$\begin{aligned} \Psi_4^{\text{off}} &= \frac{i}{2} \sigma_3 (\Psi_{3,x}^{\text{off}} - [Q, \Psi_3^{\text{diag}}]) = \frac{i}{2} \sigma_3 \Psi_{3,x}^{\text{off}} \\ &= \frac{i}{2} \sigma_3 \partial_x \left( -\frac{i}{4} Q_{xx} + \frac{i}{2} Q^3 \right) \\ &= \frac{i}{2} \sigma_3 \left( -\frac{i}{4} Q_{xxx} + \frac{i}{2} (Q^3)_x \right) \\ &= \frac{1}{8} \sigma_3 Q_{xxx} - \frac{1}{4} \sigma_3 (Q^3)_x. \end{aligned}$$

Since  $Q_x Q = Q Q_x$  for the present  $Q$  given in (1.3), we have

$$(Q^3)_x = Q_x Q^2 + Q Q_x Q + Q^2 Q_x = 3Q_x Q^2,$$

and therefore

$$\Psi_4^{\text{off}} = \frac{1}{8}\sigma_3 Q_{xxx} - \frac{3}{4}\sigma_3 Q_x Q^2.$$

For the diagonal part, only the nonvanishing combinations contribute; terms involving  $\Psi_1^{\text{diag}}$  or  $\Psi_3^{\text{diag}}$  are zero and omitted. Hence

$$\begin{aligned} \Psi_4^{\text{diag}} &= -\frac{1}{2}\sigma_3 \left( \Psi_1^{\text{off}} \Psi_3^{\text{off}} + \Psi_3^{\text{off}} \Psi_1^{\text{off}} + \Psi_2^{\text{off}} \Psi_2^{\text{off}} + \Psi_2^{\text{diag}} \Psi_2^{\text{diag}} \right) \\ &= -\frac{1}{2}\sigma_3 \left( (iQ) \left( -\frac{i}{4}Q_{xx} + \frac{i}{2}Q^3 \right) + \left( -\frac{i}{4}Q_{xx} + \frac{i}{2}Q^3 \right) (iQ) \right. \\ &\quad \left. + \left( -\frac{1}{2}\sigma_3 Q_x \right)^2 + \left( \frac{1}{2}\sigma_3 Q^2 \right)^2 \right) \\ &= -\frac{1}{8}\sigma_3 (Q Q_{xx} + Q_{xx} Q - 3Q^4 - Q_x^2). \end{aligned}$$

Since  $(-i\lambda\Psi)_+ = -i\lambda\sigma_3 + Q$ , we obtain

$$\Phi_x \Phi^{-1} = U = (-i\lambda\Psi)_+.$$

Here the subscript  $(\cdot)_+$  denotes the projection onto the nonnegative powers of  $\lambda$  in the Laurent expansion. This motivates the definition of the AKNS hierarchy by prescribing the time evolution

$$(1.18) \quad \Phi_{t_n} = (-i\lambda^n \Psi)_+ \Phi \equiv V_n \Phi.$$

The first few members of the hierarchy are given by

$$\begin{aligned} n = 2 &\Rightarrow \text{nonlinear Schrödinger (NLS) flow,} \\ n = 3 &\Rightarrow \text{complex mKdV flow,} \Rightarrow \text{mKdV flow.} \end{aligned}$$

In terms of the expansion coefficients  $\Psi_j$ , the operators  $V_n$  take the form

$$\begin{aligned} V_2 &= -i(\lambda^2 \Psi_0 + \lambda \Psi_1 + \Psi_2), \\ V_3 &= -i(\lambda^3 \Psi_0 + \lambda^2 \Psi_1 + \lambda \Psi_2 + \Psi_3). \end{aligned}$$

The zero-curvature equation  $U_t - V_x + [U, V] = 0$  with  $V_{\text{mKdV}} = 4V_3$  gives, by collecting the constant terms in  $\lambda$  (i.e., at order  $O(1)$ ),

$$Q_t + 4i\Psi_{3,x} + [Q, -4i\Psi_3] = 0.$$

Taking the off-diagonal part gives  $Q_t + 4i\Psi_{3,x}^{\text{off}} = 0$ , that is,

$$(1.19) \quad Q_t + 4i \left( -\frac{i}{4}Q_{xx} + \frac{i}{2}Q^3 \right)_x = Q_t + Q_{xxx} - 2(Q^3)_x = 0.$$

Since

$$Q = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \quad Q^2 = -u^2 \mathbb{I},$$

this reduces to the scalar equation

$$u_t + u_{xxx} + 2(u^3)_x = 0,$$

which is precisely the mKdV equation (1.1).

### Supplementary computations:

Using the definitions (1.4) and (1.6) we have

$$(1.20) \quad \Psi = \Phi \sigma_3 \Phi^{-1}.$$

Differentiating  $\Psi$  with respect to  $t_n$  and using (1.18) gives

$$\begin{aligned} \Psi_{t_n} &= \Phi_{t_n} \sigma_3 \Phi^{-1} - \Phi \sigma_3 \Phi^{-1} \Phi_{t_n} \Phi^{-1} \\ &= [\Phi_{t_n} \Phi^{-1}, \Psi] = [V_n, \Psi]. \end{aligned}$$

Thus

$$(1.21) \quad \Psi_{t_n} = [V_n, \Psi].$$

Next we simplify the zero-curvature equation for the pair  $(U, V_n)$ . The zero-curvature equation is

$$U_{t_n} - (V_n)_x + [U, V_n] = 0,$$

and since  $U = -i\lambda\sigma_3 + Q$  we have  $U_{t_n} = Q_{t_n}$ . With the normalization

$$V_n = \alpha_n (-i\lambda^n \Psi)_+$$

we obtain

$$Q_{t_n} = (V_n)_x - [U, V_n] = \alpha_n (-i\lambda^n \Psi_x)_+ - [U, \alpha_n (-i\lambda^n \Psi)_+].$$

Pulling out constants and using  $\Psi_x = [U, \Psi]$  gives

$$(1.22) \quad Q_{t_n} = -i\alpha_n \left( (\lambda^n [U, \Psi])_+ - [U, (\lambda^n \Psi)_+] \right).$$

Write  $U = -i\lambda\sigma_3 + Q$  and split the first term in (1.22):

$$\lambda^n [U, \Psi] = -i\lambda^{n+1} [\sigma_3, \Psi] + \lambda^n [Q, \Psi].$$

Because  $Q$  does not depend on  $\lambda$  we have

$$(\lambda^n [Q, \Psi])_+ = [Q, (\lambda^n \Psi)_+],$$

so the  $Q$ -commutator contributions cancel in (1.22). The remaining terms therefore come only from the  $\sigma_3$ -commutator:

$$\begin{aligned} Q_{t_n} &= -i\alpha_n \left( -i(\lambda^{n+1} [\sigma_3, \Psi])_+ + i\lambda [\sigma_3, (\lambda^n \Psi)_+] \right) \\ &= \alpha_n \left( \lambda [\sigma_3, (\lambda^n \Psi)_+] - (\lambda^{n+1} [\sigma_3, \Psi])_+ \right). \end{aligned}$$

To simplify the difference of finite sums, use the expansions

$$\Psi = \sum_{j \geq 0} \Psi_j \lambda^{-j}, \quad (\lambda^n \Psi)_+ = \sum_{j=0}^n \lambda^{n-j} \Psi_j.$$

A direct comparison of coefficients shows

$$\begin{aligned} \lambda [\sigma_3, (\lambda^n \Psi)_+] - (\lambda^{n+1} [\sigma_3, \Psi])_+ &= \sum_{j=0}^n \lambda^{n+1-j} [\sigma_3, \Psi_j] - \sum_{j=0}^{n+1} \lambda^{n+1-j} [\sigma_3, \Psi_j] \\ &= -\lambda^{n+1-(n+1)} [\sigma_3, \Psi_{n+1}] \\ &= -[\sigma_3, \Psi_{n+1}]. \end{aligned}$$

Hence the evolution of  $Q$  is

$$(1.23) \quad Q_{t_n} = -\alpha_n [\sigma_3, \Psi_{n+1}].$$

Since diagonal parts commute with  $\sigma_3$  and the commutator doubles the off-diagonal part, we may equivalently write

$$(1.24) \quad Q_{t_n} = -2\alpha_n \sigma_3 \Psi_{n+1}^{\text{off}}.$$

**Objectives:** Construct the  $L$  matrix for mKdV, establish its symmetries and conserved determinant, and relate the eigen-data to the associated hyperelliptic curve and Riemann–Theta functions.

Recall the AKNS spectral problem (1974)

$$(2.1) \quad \Phi_x = U(\lambda; Q) \Phi, \quad U(\lambda; Q) = -i\lambda\sigma_3 + Q,$$

where  $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$  and  $Q$  is off-diagonal. In what follows we set

$$(2.2) \quad Q = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}.$$

Introduce the gauge

$$(2.3) \quad \Phi = m(x; \lambda) e^{-i\lambda x \sigma_3}, \quad m = \mathbb{I} + \lambda^{-1}m_1 + \lambda^{-2}m_2 + O(\lambda^{-3}),$$

$$(2.4) \quad \Psi := m \sigma_3 m^{-1} = \Psi_0 + \Psi_1 \lambda^{-1} + \Psi_2 \lambda^{-2} + \dots,$$

so that  $\Psi^2 = \mathbb{I}$  and  $\Psi$  satisfies the stationary zero-curvature equation

$$(2.5) \quad \Psi_x = [U, \Psi].$$

Decompose a matrix  $A$  into diagonal and off-diagonal parts  $A = A^{\text{diag}} + A^{\text{off}}$ . From (2.5) and  $\Psi^2 = \mathbb{I}$  one obtains the recursion

$$(2.6) \quad \Psi_{j+1}^{\text{off}} = \frac{i}{2}\sigma_3 \left( \Psi_{j,x}^{\text{off}} - [Q, \Psi_j^{\text{diag}}] \right), \quad j \geq 0,$$

$$(2.7) \quad \Psi_{j+1}^{\text{diag}} = -\frac{1}{2}\sigma_3 \sum_{k=1}^j \left( \Psi_k^{\text{diag}} \Psi_{j-k+1}^{\text{diag}} + \Psi_k^{\text{off}} \Psi_{j-k+1}^{\text{off}} \right), \quad j \geq 1.$$

Hence each  $\Psi_j$  is a differential polynomial in  $Q$  and its  $x$ -derivatives.

One checks that

$$(2.8) \quad \Phi_x \Phi^{-1} = m_x m^{-1} - i\lambda m \sigma_3 m^{-1} = -i(\lambda \Psi)_+.$$

Let  $\Phi(x, t_n; \lambda) = m(x, t_n; \lambda) e^{-i(\lambda x + \lambda^n t_n) \sigma_3}$ , then

$$(2.9) \quad \Phi_{t_n} \Phi^{-1} = m_{t_n} m^{-1} - i\lambda^n m \sigma_3 m^{-1} = -i(\lambda^n \Psi)_+ \equiv V_n,$$

so that  $V_n = -i(\lambda^n \Psi)_+$  and

$$(2.10) \quad \Psi_{t_n} = [V_n, \Psi].$$

The compatibility  $\Phi_{xt_n} = \Phi_{t_n x}$  is equivalent to the zero curvature equation

$$(2.11) \quad U_{t_n} - V_{n,x} + [U, V_n] = 0.$$

**Definition 1.** Let  $g \geq 0$  be a fixed integer, and let  $\{\alpha_j\}_{j=0}^{g+1}$  be constants. Define

$$(2.12) \quad L := -i \sum_{j=0}^{g+1} \alpha_j (\lambda^j \Psi(x, t, \lambda))_+,$$

where  $(\cdot)_+$  denotes the projection onto nonnegative powers of  $\lambda$ .

It is convenient to rewrite  $L$  by expanding the nonnegative part explicitly. Using  $\Psi = \sum_{k \geq 0} \Psi_k \lambda^{-k}$  we have

$$(\lambda^j \Psi)_+ = \sum_{k=0}^j \Psi_k \lambda^{j-k},$$

hence

$$(2.13) \quad L(\lambda) = -i \sum_{j=0}^{g+1} \alpha_j \sum_{k=0}^j \Psi_k \lambda^{j-k} = -i \sum_{l=0}^{g+1} \left( \sum_{j=l}^{g+1} \alpha_j \Psi_{j-l} \right) \lambda^l,$$

where we set  $l = j - k$  to obtain the last form.

**Proposition 1.** Suppose  $L$  as defined in (2.12) satisfies the stationary zero-curvature equations

$$(2.14) \quad L_x = [U, L], \quad L_{t_n} = [V_n, L].$$

Then the following additional constraint holds:

$$(2.15) \quad \sum_{j=1}^{g+1} \alpha_j \Psi_{j+1}^{\text{off}}(x, t_n) + i \alpha_0 Q = 0.$$

*Proof.* Set

$$(2.16) \quad \widehat{L}(\lambda) := -i \left( \sum_{j=0}^{g+1} \alpha_j \lambda^j \right) \Psi(\lambda).$$

By construction  $\widehat{L}_x = [U, \widehat{L}]$ . Expand  $\Psi = \sum_{k \geq 0} \Psi_k \lambda^{-k}$  and substitute into  $\widehat{L}$ :

$$(2.17) \quad \widehat{L}(\lambda) = -i \sum_{j=0}^{g+1} \sum_{k \geq 0} \alpha_j \Psi_k \lambda^{j-k} = -i \sum_{l=-\infty}^{g+1} \left( \sum_{j=0}^{g+1} \alpha_j \Psi_{j-l} \right) \lambda^l,$$

where we set  $l = j - k$ . Decompose  $\widehat{L} = \widehat{L}_+ + \widehat{L}_-$  into nonnegative and negative powers of  $\lambda$ . From  $\widehat{L}_x = [U, \widehat{L}]$  we obtain

$$\left( \widehat{L}_+ + \widehat{L}_- \right)_x = [U, \widehat{L}_+ + \widehat{L}_-].$$

After extracting nonnegative powers of  $\lambda$ , we have

$$(\widehat{L}_+)_x - [U, \widehat{L}_+] = \left( [-i\lambda\sigma_3, \widehat{L}_-] \right)_+ = -i \left[ \sigma_3, \sum_{j=0}^{g+1} \alpha_j \Psi_{j+1} \right] = -2i\sigma_3 \sum_{j=0}^{g+1} \alpha_j \Psi_{j+1}^{\text{off}} = 0,$$

which yields the constraint (2.15).

Next, to ensure compatibility, we require to prove  $L_{xt_n} = L_{t_n x}$ . Observe that

$$\begin{aligned} L_{xt_n} &= [U_{t_n}, L] + [U, [V_n, L]], \\ L_{t_n x} &= [V_{n,x}, L] + [V_n, [U, L]]. \end{aligned}$$

Applying the Jacobi identity,

$$[[A, B], C] + [[C, A], B] + [[B, C], A] = 0,$$

we find

$$[U, [V_n, L]] - [V_n, [U, L]] = [[L, V_n], U] + [[U, L], V_n] = [[U, V_n], L].$$

Therefore, it follows from the zero-curvature condition (2.11) that

$$L_{xt_n} - L_{t_n x} = [U_{t_n} - V_{n,x} + [U, V_n], L] = 0.$$

□

Specializing to genus  $g = 2$ , define

$$(2.18) \quad L(x, t; \lambda) = \frac{1}{4} V(\lambda; Q) + \alpha_1 U(\lambda; Q),$$

$$(2.19) \quad V(\lambda; Q) = 4\lambda^2 U(\lambda; Q) + 2i\lambda\sigma_3(Q_x - Q^2) - (Q_{xx} - 2Q^3).$$

The zero curvature condition for the mKdV equation is

$$U_t - V_x + [U, V] = 0,$$

which yields the scalar modified KdV equation

$$(2.20) \quad u_t + u_{xxx} + 6u^2 u_x = 0, \quad u(x, t) \in \mathbb{R}.$$

The matrices  $U$  and  $V$  satisfy the following symmetries:

$$(2.21) \quad \begin{aligned} U^\dagger(\lambda^*; Q) &= -U(\lambda; Q), & U^\top(-\lambda; Q) &= -U(\lambda; Q), \\ V^\dagger(\lambda^*; Q) &= -V(\lambda; Q), & V^\top(-\lambda; Q) &= -V(\lambda; Q). \end{aligned}$$

As a consequence the matrix  $L$  inherits analogous symmetries:

$$(2.22) \quad L^\dagger(x, t; \lambda^*) = -L(x, t; \lambda), \quad L^\top(x, t; -\lambda) = -L(x, t; \lambda).$$

**Proposition 2.** *The determinant of  $L$  is independent of  $x$  and  $t$ :*

$$\partial_x \det L = \partial_t \det L = 0.$$

*Proof.* By Jacobi's formula (or Abel's theorem applying to  $L_x = (L_x L^{-1})L$ ), for any differentiable matrix  $L(x, t; \lambda)$

$$\partial_x \det L = \det L \operatorname{tr}(L_x L^{-1}).$$

Using the stationary zero-curvature equation  $L_x = [U, L] = UL - LU$  we obtain

$$\partial_x \det L = \det L \operatorname{tr}((UL - LU)L^{-1}) = \det L (\operatorname{tr} U - \operatorname{tr}(L^{-1}LU)) = 0,$$

since  $\operatorname{tr}(LUL^{-1}) = \operatorname{tr}(L^{-1}LU)$ . The identical argument with  $\partial_t L = [V, L]$  yields  $\partial_t \det L = 0$ . Hence  $\det L$  does not depend on  $x$  or  $t$ .  $\square$

Therefore the determinant of  $L$  is constant in  $x$  and  $t$ . Write its factorization as

$$(2.23) \quad \det L(\lambda) = \prod_{j=1}^6 (\lambda - \hat{\lambda}_j).$$

By the symmetry relations (2.22) each root  $\hat{\lambda}_j$  forces  $\hat{\lambda}_j^*$  and  $-\hat{\lambda}_j$  to be roots as well. Two possible root structures follow.

**Case 1.** Three purely imaginary pairs:

$$(2.24) \quad \det L(\lambda) = (\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2), \quad \lambda_j \in i\mathbb{R}.$$

**Case 2.** One nonreal quadruple together with one purely imaginary pair:

$$(2.25) \quad \det L(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_1^*)(\lambda + \lambda_1)(\lambda + \lambda_1^*)(\lambda - \lambda_2)(\lambda + \lambda_2),$$

with  $\lambda_1 \in \mathbb{C} \setminus (\mathbb{R} \cup i\mathbb{R})$  and  $\lambda_2 \in i\mathbb{R}$ .

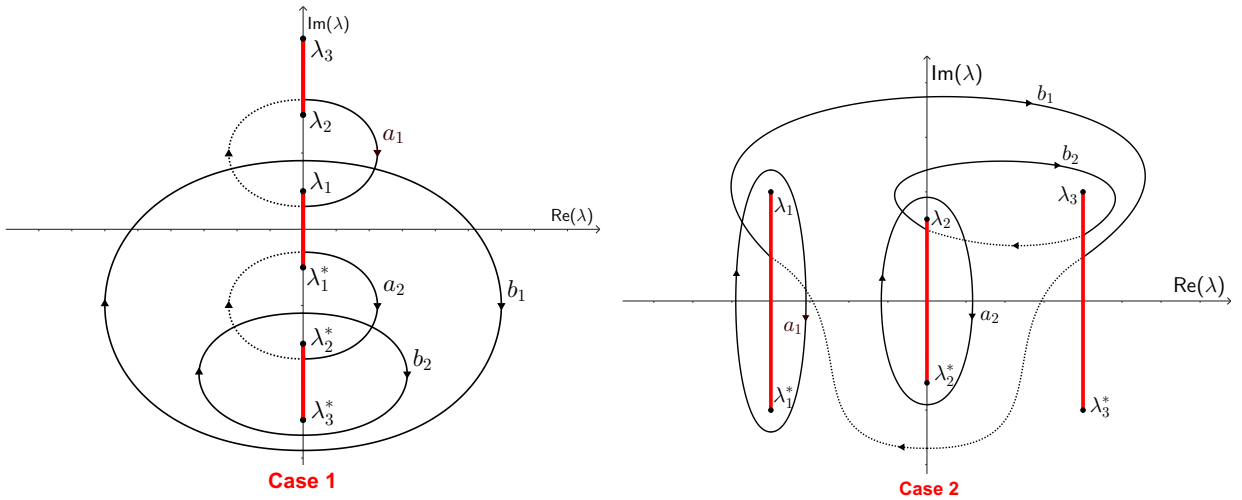


FIGURE 1. The homology basis for the curve  $y^2 = \prod_{i=1}^3 (\lambda^2 - \lambda_i^2)$ .



The matrix elements of  $L$  take the explicit form

$$\begin{aligned}
(2.26) \quad L_{11} &= -i\lambda^3 - i\lambda\left(\alpha_1 - \frac{1}{2}u^2\right), \quad L_{22} = -L_{11}, \\
L_{12} &= u\left(\lambda^2 + i\frac{u_x}{2u}\lambda - \frac{u_{xx}}{4u} + \alpha_1 - \frac{1}{2}u^2\right) = u(\lambda - \mu_1)(\lambda - \mu_2), \\
L_{21} &= -u\left(\lambda^2 - i\frac{u_x}{2u}\lambda - \frac{u_{xx}}{4u} + \alpha_1 - \frac{1}{2}u^2\right) = -u(\lambda - \mu_1^*)(\lambda - \mu_2^*),
\end{aligned}$$

where  $\mu_1, \mu_2$  denote the (in general complex) roots of the quadratic factor in  $L_{12}/u$ . The conjugation in  $L_{21}$  is consistent with the reality of  $u$  and the symmetry relations (2.22).

Recall that

$$\begin{cases} \Phi_x = U\Phi, \\ \Phi_t = V\Phi, \end{cases} \quad \begin{cases} L_x = [U, L], \\ L_t = [V, L], \end{cases}$$

It follows that  $L\Phi$  also satisfies the same Lax pair, since

$$\begin{aligned}
(L\Phi)_x &= L_x\Phi + L\Phi_x = (UL - LU)\Phi + LU\Phi = U(L\Phi), \\
(L\Phi)_t &= V(L\Phi).
\end{aligned}$$

This observation is also a direct consequence of the identity  $L = \Psi = \Phi\sigma_3\Phi^{-1}$  which implies  $L\Phi = \Phi\sigma_3$ .

Assume  $\Phi(x, t; \lambda)$  is normalized by  $\Phi(0, 0; \lambda) = \mathbb{I}$ . By uniqueness of solutions of the linear ODEs we obtain

$$(2.27) \quad L(x, t; \lambda) \Phi(x, t; \lambda) = \Phi(x, t; \lambda) L(0, 0; \lambda).$$

In particular, evaluation at  $(x, t) = (0, 0)$  yields  $L(0, 0; \lambda)$ . So knowledge of  $L(0, 0; \lambda)$  determines the action of  $L$  on the columns of  $\Phi$  for all  $(x, t)$ .

Set

$$(2.28) \quad \det(L(x, t; \lambda)) = \det(L(0, 0; \lambda)) = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2) = y^2, \quad y = \pm \sqrt{\prod_{j=1}^3 (\lambda^2 - \lambda_j^2)}.$$

A short calculation shows that  $\pm iy$  are eigenvalues of  $L(0, 0; \lambda)$ :

$$\det(\pm iy \mathbb{I} - L(0, 0; \lambda)) = -y^2 - L_{11}^2 - L_{12}L_{21} = -y^2 + \det(L(0, 0; \lambda)) = 0,$$

where  $L_{ij}$  denotes the  $(i, j)$ -entry of  $L(0, 0; \lambda)$ .

Hence there exists an invertible matrix  $M$ , independent of  $x$  and  $t$ , such that

$$(2.29) \quad L(0, 0; \lambda) = M \begin{pmatrix} iy & 0 \\ 0 & -iy \end{pmatrix} M^{-1}.$$

Combining (2.27) and (2.29) yields

$$L(x, t; \lambda) \Phi(x, t; \lambda) M = \Phi(x, t; \lambda) M \begin{pmatrix} iy & 0 \\ 0 & -iy \end{pmatrix}.$$

Write the two columns of  $\Phi(x, t; \lambda)M$  as

$$\Phi(x, t; \lambda)M = (\Phi_1, \Phi_2),$$

so that

$$L(x, t; \lambda)\Phi_1 = iy\Phi_1, \quad L(x, t; \lambda)\Phi_2 = -iy\Phi_2.$$

Equivalently,

$$(L - iy)\Phi_1 = 0, \quad (L + iy)\Phi_2 = 0,$$

and therefore

$$\ker(L \mp iy) = \text{span}\{\Phi_{1,2}\}.$$

Since each eigenvector may be rescaled, we may write

$$\Phi_1 = \begin{pmatrix} \Phi_{11} \\ \Phi_{21} \end{pmatrix} = \Phi_{11} \begin{pmatrix} 1 \\ r_1 \end{pmatrix}, \quad \Phi_2 = \Phi_{12} \begin{pmatrix} 1 \\ r_2 \end{pmatrix},$$

so

$$\ker(L \mp iy) = \text{span} \begin{pmatrix} 1 \\ r_{1,2} \end{pmatrix}.$$

Writing the eigenvalue equation explicitly gives

$$\begin{pmatrix} L_{11} \mp iy & L_{12} \\ L_{21} & -L_{11} \mp iy \end{pmatrix} \begin{pmatrix} 1 \\ r_{1,2} \end{pmatrix} = 0,$$

from which one obtains the two equivalent expressions for the ratio  $r_{1,2}$ :

$$(2.30) \quad r_{1,2} = \frac{\pm iy - L_{11}}{L_{12}} = \frac{L_{21}}{L_{11} \pm iy}.$$

The point  $P = (\lambda, y)$  lies on the hyperelliptic curve (2.28), which is a compact Riemann surface of genus  $g = 2$  with punctures  $\infty^\pm$ . Here  $P \rightarrow \infty^\pm \Leftrightarrow \lambda \rightarrow \infty, y \rightarrow \pm \lambda^3$ . Denote

$$\begin{aligned} \mu &= \{\hat{\mu}_1, \hat{\mu}_2\}, & \hat{\mu}_j &= (\mu_j, iL_{11}(\mu_j)), & j &= 1, 2, \\ \mu^* &= \{\hat{\mu}_1^*, \hat{\mu}_2^*\}, & \hat{\mu}_j^* &= (\mu_j^*, -iL_{11}(\mu_j^*)), & j &= 1, 2. \end{aligned}$$

The divisor of  $r_1$  is then given by

$$\text{Div}(r_1(x, t; P)) = D_{\infty^-, \mu^*}(\lambda) - D_{\infty^+, \mu}(\lambda).$$

That is,  $r_1(x, t; P)$  has zeros at the points of  $\mu^*$  and  $\infty^-$ , and poles at the points of  $\mu$  and  $\infty^+$ . This description of the divisor is central in constructing Baker–Akhiezer functions on the hyperelliptic Riemann surface defined by

$$y^2 = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2).$$

**Definition 2** (Riemann–Theta function). *Let  $R_g$  be a compact Riemann surface of genus  $g$ . Fix a canonical homology basis and let  $B$  denote the associated Riemann period matrix, which is symmetric with negative definite real part. The Riemann–Theta function is the entire function*

$$\Theta(B; z) = \sum_{n \in \mathbb{Z}^g} \exp \left\{ \frac{1}{2} n^\top B n + n^\top z \right\},$$

where

$$n = \begin{pmatrix} n_1 \\ \vdots \\ n_g \end{pmatrix}, \quad n_j \in \mathbb{Z}, \quad z = \begin{pmatrix} z_1 \\ \vdots \\ z_g \end{pmatrix}, \quad z_j \in \mathbb{C}.$$

**Proposition 3.** *The Riemann–Theta function satisfies:*

(1) *Periodicity in the standard lattice directions:*

$$\Theta(z + 2\pi i e_k) = \Theta(z), \quad k = 1, \dots, g.$$

(2) *Quasi-periodicity in the period matrix directions:*

$$\Theta(z + B e_k) = \Theta(z) e^{-z_k - B_{kk}/2}, \quad k = 1, \dots, g.$$

(3) *Evenness:*

$$\Theta(-z) = \Theta(z).$$

*Proof.* (1) For  $z \mapsto z + 2\pi i e_k$ , the exponential in the definition acquires a factor

$$\exp(n^\top (2\pi i e_k)) = \exp(2\pi i n_k) = 1,$$

since  $n_k \in \mathbb{Z}$ . Thus  $\Theta$  is periodic in this direction.

(2) For  $z \mapsto z + B e_k$ , we compute

$$\begin{aligned}\Theta(z + B e_k) &= \sum_{n \in \mathbb{Z}^g} \exp\left\{\frac{1}{2} n^\top B n + n^\top (z + B e_k)\right\} \\ &= \sum_{n \in \mathbb{Z}^g} \exp\left\{\frac{1}{2} n^\top B n + n^\top z + n^\top B e_k\right\}.\end{aligned}$$

Write  $m = n + e_k$ . Then  $n = m - e_k$ , and the exponent becomes

$$\begin{aligned}&\frac{1}{2}(m - e_k)^\top B(m - e_k) + (m - e_k)^\top z + (m - e_k)^\top B e_k \\ &= \frac{1}{2}(m^\top B m - m^\top B e_k - e_k^\top B m + e_k^\top B e_k) + m^\top z - e_k^\top z + m^\top B e_k - e_k^\top B e_k \\ &= \frac{1}{2}m^\top B m + m^\top z - \frac{1}{2}(m^\top B e_k + e_k^\top B m) + m^\top B e_k - \frac{1}{2}e_k^\top B e_k - e_k^\top z.\end{aligned}$$

Since  $B = B^\top$ , we have  $m^\top B e_k = e_k^\top B m$ ; moreover,  $e_k^\top z = z_k$  and  $e_k^\top B e_k = B_{kk}$ . Hence

$$\frac{1}{2}(m - e_k)^\top B(m - e_k) + (m - e_k)^\top z + (m - e_k)^\top B e_k = \frac{1}{2}m^\top B m + m^\top z - z_k - \frac{1}{2}B_{kk}.$$

Therefore,

$$\Theta(z + B e_k) = e^{-z_k - B_{kk}/2} \Theta(z).$$

(3) Finally, since the defining series is invariant under  $n \mapsto -n$ , it follows that

$$\Theta(-z) = \Theta(z).$$

□

### Supplementary computations:

Since

$$U(\lambda; Q) = -i\lambda\sigma_3 + Q = \begin{pmatrix} -i\lambda & u \\ -u & i\lambda \end{pmatrix}, \quad Q = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix},$$

we have

$$Q^2 = -u^2 \mathbb{I}, \quad Q^3 = -u^2 Q.$$

Compute each term in

$$V(\lambda; Q) = 4\lambda^2 U + 2i\lambda\sigma_3(Q_x - Q^2) - (Q_{xx} - 2Q^3).$$

First,

$$4\lambda^2 U = \begin{pmatrix} -4i\lambda^3 & 4\lambda^2 u \\ -4\lambda^2 u & 4i\lambda^3 \end{pmatrix}, \quad Q_x - Q^2 = \begin{pmatrix} u^2 & u_x \\ -u_x & u^2 \end{pmatrix}.$$

Hence

$$2i\lambda\sigma_3(Q_x - Q^2) = \begin{pmatrix} 2i\lambda u^2 & 2i\lambda u_x \\ 2i\lambda u_x & -2i\lambda u^2 \end{pmatrix}.$$

Since  $Q_{xx} - 2Q^3 = Q_{xx} + 2u^2 Q$ ,

$$-(Q_{xx} - 2Q^3) = \begin{pmatrix} 0 & -(u_{xx} + 2u^3) \\ u_{xx} + 2u^3 & 0 \end{pmatrix}.$$

Adding all contributions gives

$$V(\lambda; Q) = \begin{pmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{pmatrix},$$

where

$$(2.31) \quad V_{11} = -4i\lambda^3 + 2i\lambda u^2, \quad V_{22} = -V_{11},$$

$$(2.32) \quad V_{12} = 4\lambda^2 u + 2i\lambda u_x - (u_{xx} + 2u^3),$$

$$(2.33) \quad V_{21} = -4\lambda^2 u + 2i\lambda u_x + (u_{xx} + 2u^3).$$

### Week 3: Algebro-Geometric Solutions for the mKdV Equation (3)

2025-09-28

**Objectives:** Construct algebro-geometric mKdV solutions on a genus-two curve by expressing  $r_1$  and  $u$  in Riemann theta functions and matching their asymptotics at  $\infty^\pm$ , poles and zeroes.

The Riemann–Roch theorem provides a fundamental relation between the divisor of a meromorphic function and the corresponding Abel map data on a Riemann surface. In the present setting, for the meromorphic function  $r_1(x, t; P)$  we have

$$(3.1) \quad \text{Div}(r_1(x, t; P)) = D_{\infty^-, \mu^*}(\lambda) - D_{\infty^+, \mu}(\lambda),$$

which identifies the divisor of  $r_1$  as the difference between two effective divisors supported at the points  $\infty^\pm$  and at the divisor of spectral data  $\mu, \mu^*$ .

A concrete representation of the function  $r_1$  is given in terms of Riemann theta functions:

$$(3.2) \quad r_1(x, t; P) = \hat{r}_1(x, t) \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(P) - \alpha_{P_0}(D_{\mu^*}))}{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(P) - \alpha_{P_0}(D_{\mu}))} \exp(\Omega_1(P) - \ln(\omega_1)), \quad P = (\lambda, y).$$

Here  $\Theta$  denotes the Riemann theta function,  $\mathcal{C}$  is the vector of Riemann constants,  $\mathcal{A}_{P_0}$  is the Abel map based at  $P_0$ , and  $\alpha_{P_0}$  is its linear extension to divisors. The exponential factor is determined by an Abelian integral  $\Omega_1$ , while  $\hat{r}_1(x, t)$  is a normalizing prefactor independent of  $\lambda$ .

The underlying algebraic curve  $R_2$  associated with this construction is the genus-two hyperelliptic Riemann surface

$$(3.3) \quad R_2 := \{(\lambda, y) \mid y^2 = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2)\}, \quad P \rightarrow \infty^\pm \Leftrightarrow \lambda \rightarrow \infty, y \rightarrow \pm \lambda^3 + O(\lambda).$$

Thus each point  $P \in R_2$  is represented by  $(\lambda, y)$ , where

$$y = \pm \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}.$$

To construct the Jacobi variety one chooses a canonical homology basis  $\{a_1, a_2; b_1, b_2\}$  on  $R_2$ , satisfying

$$(3.4) \quad a_i \circ a_j = 0, \quad b_i \circ b_j = 0, \quad a_i \circ b_j = -b_j \circ a_i = \delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases} \quad i, j = 1, 2.$$

A basis of holomorphic differentials is given by

$$(3.5) \quad \oint_{a_j} w_i d\lambda = 2\pi i \delta_{ij}, \quad \oint_{b_j} w_i d\lambda = B_{ij}, \quad w_i(\lambda) = \sum_{k=0}^1 d_{ik} \lambda^k y^{-1}, \quad d_{ik} \in \mathbb{C},$$

where the normalization guarantees the  $a$ -periods form the identity matrix and the  $b$ -periods form the Riemann period matrix  $B = (B_{ij})$ .

The Abel map is defined using these normalized differentials. First, introduce the period lattice

$$(3.6) \quad L_2 = \{z \in \mathbb{C}^2 \mid z = 2\pi i(n + Bm), \quad n, m \in \mathbb{Z}^2\},$$

where  $B$  is the period matrix and  $n, m, z$  are column vectors in  $\mathbb{C}^2$ . The Jacobian of  $R_2$  is then

$$(3.7) \quad J(R_2) = \mathbb{C}^2 / L_2.$$

For a base point  $P_0 \in R_2$ , the Abel map sends a point  $P \in R_2$  to

$$(3.8) \quad \mathcal{A}_{P_0} : R_2 \rightarrow J(R_2),$$

$$(3.9) \quad P \mapsto \mathcal{A}_{P_0}(P) = (\mathcal{A}_{P_0,1}(P), \mathcal{A}_{P_0,2}(P)) = \left( \int_{P_0}^P w_1 d\lambda, \int_{P_0}^P w_2 d\lambda \right).$$

This definition extends linearly to divisors. A divisor  $D = \sum_{P \in R_2} D(P) \cdot P$  is a finite formal sum of points  $P$  on  $R_2$  with integer coefficients  $D(P) \in \mathbb{Z}$ , and we set

$$(3.10) \quad \alpha_{P_0}(D) = \sum_{P \in R_2} D(P) \mathcal{A}_{P_0}(P).$$

Finally, we introduce the Abelian integral

$$(3.11) \quad \Omega_1(P) = \int_{P_0}^P d\Omega_1, \quad \oint_{a_i} d\Omega_1 = 0,$$

with the prescribed asymptotics at the points at infinity,

$$(3.12) \quad \Omega_1(P) = \pm \left( \ln(\lambda) + \ln(\omega_1) + o(1) \right), \quad P \rightarrow \infty^\pm.$$

This integral provides the exponential term appearing in (3.2) and ensures the correct divisor structure of  $r_1$ .

Recall that the function  $r_1(x, t; P)$  admits the representation

$$(3.13) \quad r_1(x, t; P) = \frac{L_{21}}{iy + L_{11}},$$

where

$$(3.14) \quad L_{21} = -u \left( \lambda^2 - \frac{i u_x}{2u} \lambda - \frac{u_{xx}}{4u} + \alpha_1 - \frac{u^2}{2} \right), \quad L_{11} = -i\lambda^3 - i\lambda \left( \alpha_1 - \frac{u^2}{2} \right).$$

To analyze its asymptotic behavior, note first that on the hyperelliptic curve  $R_2$ ,

$$(3.15) \quad y = \pm \left( \lambda^3 - \frac{1}{2}(\lambda_1^2 + \lambda_2^2 + \lambda_3^2)\lambda \right) + O(\lambda^{-1}), \quad P \rightarrow \infty^\pm.$$

As  $P \rightarrow \infty^-$ , the expansion of the denominator yields

$$(3.16) \quad L_{11} + iy = -2i\lambda^3 - i\lambda \left( \alpha_1 - \frac{u^2}{2} \right) + \frac{i}{2}(\lambda_1^2 + \lambda_2^2 + \lambda_3^2)\lambda + O(\lambda^{-1}),$$

so that

$$(3.17) \quad r_1(x, t; P) = \frac{L_{21}}{iy + L_{11}} = -\frac{i u}{2\lambda} + O(\lambda^{-2}), \quad P \rightarrow \infty^-.$$

Similarly, near  $\infty^+$  one obtains

$$(3.18) \quad r_1(x, t; P) = \frac{iy - L_{11}}{L_{12}} = \frac{2i}{u}\lambda + O(1), \quad P \rightarrow \infty^+.$$

On the other hand, from the Riemann theta function representation (3.2), the asymptotics at the two points at infinity are

$$(3.19) \quad r_1(x, t; P) = \begin{cases} \hat{r}_1(x, t) \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*}))}{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu))} \lambda + O(1), & P \rightarrow \infty^+, \\ \hat{r}_1(x, t) \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}))}{\omega_1^2 \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))} \lambda^{-1} + O(\lambda^{-2}), & P \rightarrow \infty^-. \end{cases}$$

By comparing these expansions with the direct computation, one obtains the following theta-function formula for the potential  $u(x, t)$ :

$$(3.20) \quad u(x, t) = 2i \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu))}{\hat{r}_1(x, t) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*}))} = 2i \hat{r}_1(x, t) \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}))}{\omega_1^2 \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))}.$$

This identity expresses the potential  $u$  in terms of theta functions associated with the Jacobian of the spectral curve, thereby linking the algebro-geometric spectral data with the physical field variable.

Recall that the column vector

$$\Phi_1 = \begin{pmatrix} \Phi_{11} \\ \Phi_{21} \end{pmatrix} = \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix}$$

satisfies both parts of the Lax pair:

$$(3.21) \quad \partial_x \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix} = \begin{pmatrix} -i\lambda & u \\ -u & i\lambda \end{pmatrix} \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix}, \quad \partial_t \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix} = \begin{pmatrix} V_{11} & V_{12} \\ V_{21} & -V_{11} \end{pmatrix} \begin{pmatrix} \Phi_{11} \\ \Phi_{11} r_1 \end{pmatrix}.$$

Hence the scalar component  $\Phi_{11}$  satisfies

$$(3.22) \quad \Phi_{11,x} = (-i\lambda + u r_1) \Phi_{11}, \quad \Phi_{11,t} = (V_{11} + V_{12} r_1) \Phi_{11}.$$

Integrating the differential 1-form in (3.22) from the base point  $(x_0, t_0)$  to  $(x, t)$  yields

$$(3.23) \quad \Phi_{11}(x, t; \lambda) = \Phi_{11}(x_0, t_0; \lambda) \exp \left( \int_{(x_0, t_0)}^{(x, t)} \left[ (-i\lambda + u r_1) dx + (V_{11} + V_{12} r_1) dt \right] \right).$$

Hence  $\Phi_{11} \in \mathcal{M}(R_2 \setminus \{\infty^\pm\})$ , the space of functions meromorphic on  $R_2$  away from  $\infty^\pm$ . The representation (3.23) also shows that  $\Phi_{11}$  has an essential singularity at each of  $\infty^\pm$ . For convenience, we set the initial point as  $(x_0, t_0) = (0, 0)$ .

The large- $\lambda$  asymptotics differ on the two sheets. On the sheet with  $P \rightarrow \infty^-$  we fix the normalization

$$\Phi_{1,0}^- := \lim_{P \rightarrow \infty^-} \Phi_{11}(x_0, t_0; \lambda) = 1,$$

and obtain

$$(3.24) \quad \begin{aligned} \Phi_{11}(x, t; \lambda) &= \exp(-i\lambda(x + 4\lambda^2 t)) + O(1), & P \rightarrow \infty^-, \\ \Phi_{21}(x, t; \lambda) &= r_1 \Phi_{11} = O(\lambda^{-1}) \exp(-i\lambda(x + 4\lambda^2 t)), & P \rightarrow \infty^-. \end{aligned}$$

On the other sheet,  $P \rightarrow \infty^+$ , the expansion of  $r_1$  is

$$(3.25) \quad r_1(x, t; P) = \frac{iy - L_{11}}{L_{12}} = \frac{2i}{u} \lambda + \frac{u_x}{u^2} + O(\lambda^{-1}), \quad P \rightarrow \infty^+.$$

Consequently

$$(3.26) \quad -i\lambda + u r_1 = i\lambda + (\ln u)_x + O(\lambda^{-1}), \quad P \rightarrow \infty^+.$$

Substituting (3.26) into (3.23) gives, with the normalization constant

$$(3.27) \quad \Phi_{1,0}^+ := \lim_{P \rightarrow \infty^+} \Phi_{11}(x_0, t_0; \lambda),$$

the asymptotic formulas

$$(3.28) \quad \begin{aligned} \Phi_{11}(x, t; \lambda) &= \Phi_{1,0}^+ u(x, t) \exp(i\lambda(x + 4\lambda^2 t)) (1 + O(\lambda^{-1})), & P \rightarrow \infty^+, \\ \Phi_{21}(x, t; \lambda) &= r_1 \Phi_{11} = 2i\lambda \Phi_{1,0}^+ \exp(i\lambda(x + 4\lambda^2 t)) (1 + O(\lambda^{-1})), & P \rightarrow \infty^+. \end{aligned}$$

Recall

$$(3.29) \quad r_1(\lambda) = \frac{iy(\lambda) - L_{11}(\lambda)}{L_{12}(\lambda)} = \frac{iy - L_{11}}{u(\lambda - \mu_1)(\lambda - \mu_2)},$$

$$(3.30) \quad r_1(\lambda) = \frac{L_{21}(\lambda)}{iy(\lambda) + L_{11}(\lambda)} = \frac{-u(\lambda - \mu_1^*)(\lambda - \mu_2^*)}{iy + L_{11}}.$$

We write

$$L(\lambda) = \begin{pmatrix} L_{11}(\lambda) & L_{12}(\lambda) \\ L_{21}(\lambda) & -L_{11}(\lambda) \end{pmatrix}, \quad U(\lambda) = \begin{pmatrix} -i\lambda & u \\ -u & i\lambda \end{pmatrix},$$

and recall the stationary zero-curvature equation

$$(3.31) \quad L_x = [U, L].$$

With  $\sigma_3 = \text{diag}(1, -1)$  and off-diagonal part  $L^{\text{off}} = \begin{pmatrix} 0 & L_{12} \\ L_{21} & 0 \end{pmatrix}$ , we compute

$$\begin{aligned} [U, L] &= -2i\lambda \sigma_3 L^{\text{off}} + [Q, L] \\ &= -2i\lambda \sigma_3 L^{\text{off}} + \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix} \begin{pmatrix} L_{11} & L_{12} \\ L_{21} & -L_{11} \end{pmatrix} - \begin{pmatrix} L_{11} & L_{12} \\ L_{21} & -L_{11} \end{pmatrix} \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix} \\ &= -2i\lambda \begin{pmatrix} 0 & L_{12} \\ -L_{21} & 0 \end{pmatrix} + \begin{pmatrix} u(L_{21} + L_{12}) & -2uL_{11} \\ -2uL_{11} & -u(L_{21} + L_{12}) \end{pmatrix} \\ (3.32) \quad &= \begin{pmatrix} u(L_{21} + L_{12}) & -2i\lambda L_{12} - 2uL_{11} \\ 2i\lambda L_{21} - 2uL_{11} & -u(L_{21} + L_{12}) \end{pmatrix}. \end{aligned}$$

Thus from (3.31) we obtain

$$\begin{aligned} (3.33) \quad \partial_x L_{11} &= u(L_{21} + L_{12}), \\ \partial_x L_{12} &= -2i\lambda L_{12} - 2uL_{11}, \\ \partial_x L_{21} &= 2i\lambda L_{21} - 2uL_{11}. \end{aligned}$$

By assumption

$$(3.34) \quad L_{12}(\lambda) = u(\lambda - \mu_1)(\lambda - \mu_2).$$

Differentiating with respect to  $x$  gives

$$(3.35) \quad \partial_x L_{12}(\lambda) = u_x(\lambda - \mu_1)(\lambda - \mu_2) - u \mu_{1,x}(\lambda - \mu_2) - u \mu_{2,x}(\lambda - \mu_1).$$

Evaluating (3.35) at  $\lambda = \mu_1$  leaves only the middle term:

$$(3.36) \quad \partial_x L_{12}(\mu_1) = -u \mu_{1,x}(\mu_1 - \mu_2).$$

On the other hand, from (3.33) evaluated at  $\lambda = \mu_1$  we have

$$(3.37) \quad \partial_x L_{12}(\mu_1) = -2u L_{11}(\mu_1).$$

Equating the two expressions yields

$$(3.38) \quad \mu_{1,x} = \frac{2L_{11}(\mu_1)}{\mu_1 - \mu_2}.$$

Next, the determinant relation  $y^2 = \det L = -L_{11}^2 - L_{12}L_{21}$  at  $\lambda = \mu_1$  reduces to

$$(3.39) \quad y(\mu_1)^2 = -L_{11}(\mu_1)^2,$$

since  $L_{12}(\mu_1) = 0$ . Hence  $-iy(\mu_1) = \pm L_{11}(\mu_1)$ . We fix the sign by setting

$$(3.40) \quad -iy(\mu_1) = L_{11}(\mu_1),$$

so that  $P = (\mu_1, iL_{11}(\mu_1))$  represents a pole of the meromorphic function  $r_1(P) = r_1(\lambda)$  on the Riemann surface  $R_2$ .

Consider now the principal part of  $u r_1$  near  $\lambda = \mu_1$ . From (3.29) and (3.34),

$$(3.41) \quad u r_1(\lambda) = \frac{iy(\lambda) - L_{11}(\lambda)}{(\lambda - \mu_1)(\lambda - \mu_2)} = \frac{iy(\mu_1) - L_{11}(\mu_1)}{(\mu_1 - \mu_2)} \cdot \frac{1}{\lambda - \mu_1} + O(1), \quad \lambda \rightarrow \mu_1.$$

Using (3.40) and (3.38),

$$iy(\mu_1) - L_{11}(\mu_1) = -2L_{11}(\mu_1) = -(\mu_1 - \mu_2) \mu_{1,x}.$$

Thus (3.41) becomes

$$(3.42) \quad u r_1(\lambda) = -\frac{\mu_{1,x}}{\lambda - \mu_1} + O(1) = \partial_x(\ln(\lambda - \mu_1)) + O(1), \quad \lambda \rightarrow \mu_1.$$

Then by (3.23) and (3.42) we conclude that  $\lambda = \mu_1$  is a simple zero of  $\Phi_{11}(x, t; \lambda)$ .

**Supplementary computations:**

Evaluate (3.35) at  $\lambda = \mu_2$ . Only the third term survives, so

$$(3.43) \quad \partial_x L_{12}(\mu_2) = -u \mu_{2,x}(\mu_2 - \mu_1).$$

From (3.33) at  $\lambda = \mu_2$  we have

$$(3.44) \quad \partial_x L_{12}(\mu_2) = -2u L_{11}(\mu_2).$$

Equating these expressions yields

$$(3.45) \quad \mu_{2,x} = \frac{2L_{11}(\mu_2)}{\mu_2 - \mu_1}.$$

Adopting the same sheet convention as for  $\mu_1$  we take

$$(3.46) \quad -iy(\mu_2) = L_{11}(\mu_2).$$

Compute the principal part of  $u r_1$  near  $\lambda = \mu_2$ . From (3.29) and (3.34),

$$(3.47) \quad u r_1(\lambda) = \frac{iy(\lambda) - L_{11}(\lambda)}{(\lambda - \mu_1)(\lambda - \mu_2)} = \frac{iy(\mu_2) - L_{11}(\mu_2)}{(\mu_2 - \mu_1)} \cdot \frac{1}{\lambda - \mu_2} + O(1), \quad \lambda \rightarrow \mu_2.$$

Using (3.46) and (3.45) we obtain

$$iy(\mu_2) - L_{11}(\mu_2) = -2L_{11}(\mu_2) = -(\mu_2 - \mu_1) \mu_{2,x}.$$

Hence (3.47) becomes

$$(3.48) \quad u r_1(\lambda) = -\frac{\mu_{2,x}}{\lambda - \mu_2} + O(1) = \partial_x(\ln(\lambda - \mu_2)) + O(1), \quad \lambda \rightarrow \mu_2.$$

Therefore, by (3.23) and (3.48),  $\lambda = \mu_2$  is a simple zero of  $\Phi_{11}(x, t; \lambda)$ .

We next consider the time evolution governed by

$$(3.49) \quad L_t = [V, L], \quad V(\lambda) = \begin{pmatrix} V_{11}(\lambda) & V_{12}(\lambda) \\ V_{21}(\lambda) & -V_{11}(\lambda) \end{pmatrix}.$$

Direct computation of the commutator yields

$$[V, L] = VL - LV = \begin{pmatrix} V_{12}L_{21} - L_{12}V_{21} & 2V_{11}L_{12} - 2L_{11}V_{12} \\ 2L_{11}V_{21} - 2V_{11}L_{21} & L_{12}V_{21} - V_{12}L_{21} \end{pmatrix},$$

since the trace vanishes and the anti-diagonal structure pairs terms. Consequently from (3.49) we obtain the componentwise equations

$$(3.50) \quad \begin{aligned} \partial_t L_{11} &= V_{12}L_{21} - L_{12}V_{21}, \\ \partial_t L_{12} &= 2V_{11}L_{12} - 2L_{11}V_{12}, \\ \partial_t L_{21} &= 2L_{11}V_{21} - 2V_{11}L_{21}. \end{aligned}$$

By assumption

$$(3.51) \quad L_{12}(\lambda) = u(\lambda - \mu_1)(\lambda - \mu_2),$$

where  $u = u(x, t)$  and  $\mu_j = \mu_j(x, t)$ . Differentiating (3.51) with respect to  $t$  gives

$$(3.52) \quad \partial_t L_{12}(\lambda) = u_t(\lambda - \mu_1)(\lambda - \mu_2) - u \mu_{1,t}(\lambda - \mu_2) - u \mu_{2,t}(\lambda - \mu_1).$$

Evaluating (3.52) at  $\lambda = \mu_1$  annihilates the first and third terms and leaves

$$(3.53) \quad \partial_t L_{12}(\mu_1) = -u \mu_{1,t}(\mu_1 - \mu_2).$$



On the other hand, evaluating the second equation in (3.50) at  $\lambda = \mu_1$  yields

$$(3.54) \quad \partial_t L_{12}(\mu_1) = 2V_{11}(\mu_1) L_{12}(\mu_1) - 2L_{11}(\mu_1) V_{12}(\mu_1).$$

Since  $L_{12}(\mu_1) = 0$  by (3.51), the first term on the right of (3.54) vanishes and we obtain

$$(3.55) \quad \partial_t L_{12}(\mu_1) = -2L_{11}(\mu_1) V_{12}(\mu_1).$$

Equating (3.53) and (3.55) gives the evolution law for  $\mu_1$ :

$$(3.56) \quad \mu_{1,t} = \frac{2L_{11}(\mu_1) V_{12}(\mu_1)}{u(\mu_1 - \mu_2)}.$$

Consider the principal part of  $V_{12}(\lambda) r_1(\lambda)$  near  $P = (\mu_1, iL_{11}(\mu_1))$ . From (3.29) we have

$$(3.57) \quad V_{12}(\lambda) r_1(\lambda) = V_{12}(\lambda) \frac{iy(\lambda) - L_{11}(\lambda)}{u(\lambda - \mu_1)(\lambda - \mu_2)}.$$

Expanding the numerator and evaluating at  $\lambda = \mu_1$  yields the simple pole

$$(3.58) \quad V_{12}(\lambda) r_1(\lambda) = \frac{-2L_{11}(\mu_1) V_{12}(\mu_1)}{u(\mu_1 - \mu_2)} \cdot \frac{1}{\lambda - \mu_1} + O(1), \quad \lambda \rightarrow \mu_1.$$

Using the evolution law (3.56) for  $\mu_1$  this principal part becomes

$$(3.59) \quad V_{12}(\lambda) r_1(\lambda) = -\frac{\mu_{1,t}}{\lambda - \mu_1} + O(1) = \partial_t(\ln(\lambda - \mu_1)) + O(1), \quad \lambda \rightarrow \mu_1.$$

## Week 4: Algebraic-Geometric Solutions for the mKdV Equation (4)

2025-10-14

**Objectives:** Continue to construct algebraic-geometric mKdV solutions on a genus-two curve by expressing  $r_1$  and  $u$  in Riemann theta functions and matching their asymptotics at  $\infty^\pm$ , poles and zeroes.

Recall  $P = (\lambda, y)$ . The genus-two hyperelliptic curve and its points at infinity are

$$(4.1) \quad R_2 = \{(\lambda, y) \mid y^2 = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2)\}, \quad P \rightarrow \infty^\pm \iff \lambda \rightarrow \infty, y \rightarrow \pm \lambda^3.$$

Define the points

$$(4.2) \quad \mu = \{\hat{\mu}_1, \hat{\mu}_2\}, \quad \hat{\mu}_j = (\mu_j, iL_{11}(\mu_j)), \quad \mu^* = \{\hat{\mu}_1^*, \hat{\mu}_2^*\}, \quad \hat{\mu}_j^* = (\mu_j^*, -iL_{11}(\mu_j^*)),$$

so that  $\mu^*$  are the zeros and  $\mu$  the poles of  $r_1$ .

At a pole  $\hat{\mu}_1 = (\mu_1, iL_{11}(\mu_1))$  the meromorphic function  $r_1(\lambda)$  has the local expansion

$$(4.3) \quad r_1(\lambda) = \frac{iy(\lambda) - L_{11}(\lambda)}{L_{12}(\lambda)} = \frac{iy - L_{11}}{u(\lambda - \mu_1)(\lambda - \mu_2)} = \frac{-2L_{11}(\mu_1)}{u(\mu_1 - \mu_2)} \cdot \frac{1}{\lambda - \mu_1} + O(1), \quad \lambda \rightarrow \mu_1.$$

Recall that  $r_1(x, t; P)$  is defined in (3.2) and that  $\Phi_{11}(x, t; P) = \Phi_{11}(x, t; \lambda)$  is given by the integral representation (3.23). Choosing a continuous path from  $(x_0, t_0)$  to  $(x, t)$  we have

$$(4.4) \quad \begin{aligned} \Phi_{11}(x, t; \lambda) &= \Phi_{11}(x_0, t_0; \lambda) \exp \left( \int_{(x_0, t_0)}^{(x, t)} (-i\lambda + u r_1) dx_1 + (V_{11} + V_{12} r_1) dt_1 \right) \\ &= \Phi_{11}(x_0, t_0; \lambda) \exp \left( \int_{x_0}^x (-i\lambda + u r_1)(x_1, t_0) dx_1 + \int_{t_0}^t (V_{11} + V_{12} r_1)(x, t_1) dt_1 \right). \end{aligned}$$

The singularity structure of  $\Phi_{11}$  follows from the pole structure of  $r_1$ . From (4.3) and the explicit expressions for  $u r_1$  and  $V_{12} r_1$  (see (3.42) and (3.48)) one obtains, as  $\lambda \rightarrow \mu_j$ ,

$$(4.5) \quad u r_1(x_1, t_0; \lambda) = \partial_{x_1}(\ln(\lambda - \mu_j(x_1, t_0))) + O(1),$$

and

$$(4.6) \quad V_{12} r_1(x, t_1; \lambda) = \partial_{t_1}(\ln(\lambda - \mu_j(x, t_1))) + O(1).$$

Therefore each moving point  $\hat{\mu}_j(x, t) = (\mu_j(x, t), iL_{11}(\mu_j(x, t)))$  is a zero of  $\Phi_{11}(x, t; P)$ , while the initial points  $\mu_0 := \mu(x_0, t_0)$  give the poles of  $\Phi_{11}$ .

Matching the asymptotic forms (3.24) and (3.28) of  $\Phi_{11}$  as  $P \rightarrow \infty^\pm$  gives the theta-function representation

$$(4.7) \quad \Phi_{11}(x, t; P) = \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(P) - \alpha_{P_0}(D_\mu)) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}))}{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(P) - \alpha_{P_0}(D_{\mu_0})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))} e^{i(\omega_2 + \Omega_2(P))x + i(\omega_3 + \Omega_3(P))t}.$$

Here  $P_0 = (\lambda_3, 0)$ , and the Abelian integrals  $\Omega_2, \Omega_3$  are defined by

$$(4.8) \quad \Omega_2(P) := \int_{P_0}^P d\Omega_2, \quad \oint_{a_i} d\Omega_2 = 0, \quad \Omega_2(P) = \pm(\lambda + \omega_2 + o(1)), \quad P \rightarrow \infty^\pm,$$

$$(4.9) \quad \Omega_3(P) := \int_{P_0}^P d\Omega_3, \quad \oint_{a_i} d\Omega_3 = 0, \quad \Omega_3(P) = \pm(4\lambda^3 + \omega_3 + o(1)), \quad P \rightarrow \infty^\pm,$$

where  $\omega_2, \omega_3$  are constants independent of  $\lambda$ . As  $P \rightarrow \infty^-$ , (4.7) is consistent with (3.24).

As  $P \rightarrow \infty^+$ , comparing (3.28) with (4.7) yields

$$(4.10) \quad u(x, t) = \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu)) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}))}{\Phi_{1,0}^+ \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu_0})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))} e^{2i\omega_2 x + 2i\omega_3 t}.$$

Combining (3.20) and (4.10), we start from the first expression in (3.20):

$$u(x, t) = 2i \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu))}{\hat{r}_1(x, t) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*}))}.$$

Equating this with (4.10) gives

$$2i \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu))}{\hat{r}_1(x, t) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*}))} = \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu)) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}))}{\Phi_{1,0}^+ \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu_0})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))} e^{2i\omega_2 x + 2i\omega_3 t}.$$

Cancel the common factor  $\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_\mu))$  and solve for  $\hat{r}_1(x, t)$ :

$$(4.11) \quad \hat{r}_1(x, t) = \frac{2i \Phi_{1,0}^+ \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu_0})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))}{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}))} e^{-2i\omega_2 x - 2i\omega_3 t}.$$

Substitute (4.11) into the second form of (3.20),

$$u(x, t) = 2i \hat{r}_1(x, t) \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}))}{\omega_1^2 \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))}.$$

Then

$$\begin{aligned} u(x, t) &= 2i \frac{2i \Phi_{1,0}^+ \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu_0})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))}{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}))} e^{-2i\omega_2 x - 2i\omega_3 t} \\ &\quad \times \frac{\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}))}{\omega_1^2 \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))}. \end{aligned}$$

After cancelling  $\Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_\mu))$ , we obtain

$$(4.12) \quad u(x, t) = \frac{(2i)^2 \Phi_{1,0}^+ \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu_0})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu^*}))}{\omega_1^2 \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^+) - \alpha_{P_0}(D_{\mu^*})) \Theta(\mathcal{C} + \mathcal{A}_{P_0}(\infty^-) - \alpha_{P_0}(D_{\mu_0}))} e^{-2i\omega_2 x - 2i\omega_3 t}.$$

We need to determine these the parameters:

$$\alpha_{P_0}(D_\mu), \quad \alpha_{P_0}(D_{\mu^*}), \quad B, \quad w_1, w_2, \quad \omega_1, \omega_2, \omega_3, \quad \Phi_{1,0}^+.$$

The Riemann-Roch theorem gives

$$(4.13) \quad r(-D) = \deg(D) + i(D) - g + 1,$$

where

$$\mathcal{L}(D) = \{f \in \mathcal{M}(R_g) \mid (f) \geq D\}, \quad r(D) = \dim(\mathcal{L}(D)),$$

and

$$\mathcal{L}'(D) = \{\omega \in \mathcal{M}'(R_g) \mid (\omega) \geq D\}, \quad i(D) = \dim(\mathcal{L}'(D)),$$

with  $\mathcal{M}(R_g)$  the field of meromorphic functions on the genus  $g$  curve  $R_g$  and  $\mathcal{M}'(R_g)$  the space of meromorphic differentials.

Recall that

$$(4.14) \quad \text{Div}(r_1(x, t; P)) = D_{\infty^-, \mu^*}(\lambda) - D_{\infty^+, \mu}(\lambda).$$

Take the (non special) divisor  $D = D_{\infty^+, \mu}$ . Assume  $i(D) = 0$  and that  $(r_1) \geq -D_{\infty^+, \mu}$ . Then by (4.13) (with  $g = 2$  and  $\deg(D) = 3$  in the present setup)

$$(4.15) \quad r(-D_{\infty^+, \mu}) = \deg(D) - g + 1 = 3 - 2 + 1 = 2.$$

Since  $r_1$  and  $\tilde{r}_1$  have the same zeros they span a two dimensional space and hence  $r_1$  is proportional to  $\tilde{r}_1$ , i.e.

$$r_1 = c\tilde{r}_1 + d \Rightarrow r_1 = c\tilde{r}_1 \quad (c, d \in \mathbb{C}).$$

Recall the Abel map for the degree two divisor  $D_\mu = \hat{\mu}_1 + \hat{\mu}_2$  based at  $P_0$ :

$$(4.16) \quad \alpha_{P_0}(D_\mu) = \mathcal{A}_{P_0}(\hat{\mu}_1) + \mathcal{A}_{P_0}(\hat{\mu}_2) = \left( \int_{P_0}^{\hat{\mu}_1} w_1 d\lambda + \int_{P_0}^{\hat{\mu}_2} w_1 d\lambda, \int_{P_0}^{\hat{\mu}_1} w_2 d\lambda + \int_{P_0}^{\hat{\mu}_2} w_2 d\lambda \right).$$

Let  $\{w_1 d\lambda, w_2 d\lambda\}$  be a basis of holomorphic differentials normalized by

$$(4.17) \quad \oint_{a_j} w_i d\lambda = 2\pi i \delta_{ij}, \quad \oint_{b_j} w_i d\lambda = B_{ij}, \quad i, j = 1, 2.$$

On the affine chart write each differential as a rational function of  $\lambda$  divided by  $y(\lambda)$ :

$$(4.18) \quad w_i(\lambda) = (d_{i0} + d_{i1}\lambda) \frac{1}{y(\lambda)}, \quad d_{ik} \in \mathbb{C}, \quad i = 1, 2, \quad k = 0, 1.$$

Thus

$$w_1(\lambda) = \frac{d_{10} + d_{11}\lambda}{y(\lambda)}, \quad w_2(\lambda) = \frac{d_{20} + d_{21}\lambda}{y(\lambda)}.$$

Differentiate the components of  $\alpha_{P_0}(D_\mu)$  with respect to  $x$ . By the chain rule, for  $j = 1, 2$  and  $i = 1, 2$ ,

$$(4.19) \quad \frac{d}{dx} \int_{P_0}^{\hat{\mu}_j} w_i d\lambda = w_i(\hat{\mu}_j) \mu_{j,x} = \frac{d_{i0} + d_{i1}\mu_j}{y(\mu_j)} \mu_{j,x}.$$

We compute the  $w_1$ -component of  $\alpha_{P_0}(D_\mu)$ . Using (4.19) and the expressions (3.38) and (3.45) for  $\mu_{j,x}$  in terms of the Lax matrix entries (so that  $\mu_{j,x} = \frac{2L_{11}(\mu_j)}{\mu_j - \mu_k}$  with  $k \neq j$  and  $y(\mu_j) = iL_{11}(\mu_j)$ ), one obtains

$$\begin{aligned} \frac{d}{dx} \left( \int_{P_0}^{\hat{\mu}_1} (d_{10} + d_{11}\lambda) \frac{1}{y} d\lambda + \int_{P_0}^{\hat{\mu}_2} (d_{10} + d_{11}\lambda) \frac{1}{y} d\lambda \right) &= \frac{d_{10} + d_{11}\mu_1}{y(\mu_1)} \mu_{1,x} + \frac{d_{10} + d_{11}\mu_2}{y(\mu_2)} \mu_{2,x} \\ &= d_{10} \left( \frac{\mu_{1,x}}{y(\mu_1)} + \frac{\mu_{2,x}}{y(\mu_2)} \right) + d_{11} \left( \frac{\mu_1 \mu_{1,x}}{y(\mu_1)} + \frac{\mu_2 \mu_{2,x}}{y(\mu_2)} \right) \\ &= d_{10} \left( \frac{2L_{11}(\mu_1)}{\mu_1 - \mu_2} \frac{1}{iL_{11}(\mu_1)} + \frac{2L_{11}(\mu_2)}{\mu_2 - \mu_1} \frac{1}{iL_{11}(\mu_2)} \right) + d_{11} \left( \frac{2\mu_1}{i(\mu_1 - \mu_2)} + \frac{2\mu_2}{i(\mu_2 - \mu_1)} \right) \\ &= \frac{2}{i} d_{10} \left( \frac{1}{\mu_1 - \mu_2} + \frac{1}{\mu_2 - \mu_1} \right) + \frac{2}{i} d_{11} \left( \frac{\mu_1}{\mu_1 - \mu_2} + \frac{\mu_2}{\mu_2 - \mu_1} \right) \\ (4.20) \quad &= \frac{2}{i} d_{11} = -2i d_{11}. \end{aligned}$$

The  $w_2$ -component is analogous. Repeating the same substitutions gives

$$(4.21) \quad \frac{d}{dx} \left( \int_{P_0}^{\hat{\mu}_1} (d_{20} + d_{21}\lambda) \frac{1}{y} d\lambda + \int_{P_0}^{\hat{\mu}_2} (d_{20} + d_{21}\lambda) \frac{1}{y} d\lambda \right) = -2i d_{21}.$$

Combining (4.20) and (4.21) we obtain

$$(4.22) \quad \frac{d}{dx} (\alpha_{P_0}(D_\mu)) = -2i (d_{11}, d_{21}) = -i \hat{U},$$

where  $\hat{U} = (U_1, U_2)$  and  $U_i = \oint_{b_i} d\Omega_2$ .

For the time evolution induced by the higher flow one finds similarly

$$(4.23) \quad \frac{d}{dt} (\alpha_{P_0}(D_\mu)) = -i \hat{V} = -i v \hat{U} = -2i v (d_{11}, d_{21}),$$

with

$$v = 2(\lambda_1^2 + \lambda_2^2 + \lambda_3^2), \quad \hat{V} = (V_1, V_2), \quad V_i = \oint_{b_i} d\Omega_3.$$

### Supplementary computations:

Differentiate the components of  $\alpha_{P_0}(D_\mu)$  with respect to  $t$ . By the chain rule, for  $j = 1, 2$  and  $i = 1, 2$ ,

$$(4.24) \quad \frac{d}{dt} \int_{P_0}^{\hat{\mu}_j} w_i d\lambda = w_i(\hat{\mu}_j) \mu_{j,t} = \frac{d_{i0} + d_{i1}\mu_j}{y(\mu_j)} \mu_{j,t}.$$

Use the evolution law (3.56) (and the analogous formula for  $\mu_{2,t}$ ):

$$(4.25) \quad \mu_{j,t} = \frac{2L_{11}(\mu_j) V_{12}(\mu_j)}{u(\mu_j - \mu_k)}, \quad k \neq j.$$

Also use  $y(\mu_j) = iL_{11}(\mu_j)$ . Substituting these into (4.24) gives, for each  $j$ ,

$$\frac{d}{dt} \int_{P_0}^{\hat{\mu}_j} w_i d\lambda = \frac{d_{i0} + d_{i1}\mu_j}{iL_{11}(\mu_j)} \frac{2L_{11}(\mu_j) V_{12}(\mu_j)}{u(\mu_j - \mu_k)} = \frac{2}{iu} \frac{d_{i0} + d_{i1}\mu_j}{\mu_j - \mu_k} V_{12}(\mu_j).$$

Summing the contributions from  $j = 1, 2$  we obtain the two components of the time derivative of  $\alpha_{P_0}(D_\mu)$ . Thus

$$(4.26) \quad \frac{d}{dt} (\alpha_{P_0}(D_\mu)) = \frac{2}{iu} \left( \sum_{j=1}^2 \frac{d_{10} + d_{11}\mu_j}{\mu_j - \mu_k} V_{12}(\mu_j), \sum_{j=1}^2 \frac{d_{20} + d_{21}\mu_j}{\mu_j - \mu_k} V_{12}(\mu_j) \right),$$

with in each sum  $k \neq j$ .

Using the relation  $L_{12} = \frac{1}{4}V_{12} + \alpha_1 u$  from (2.18), and noting that  $L_{12}(\mu_j) = 0$  for  $j = 1, 2$ , we obtain

$$(4.27) \quad V_{12}(\mu_j) = 4(L_{12}(\mu_j) - \alpha_1 u) = -4\alpha_1 u, \quad j = 1, 2.$$

Hence for each component  $i \in \{1, 2\}$ ,

$$(4.28) \quad \sum_{j=1}^2 \frac{d_{i0} + d_{i1}\mu_j}{\mu_j - \mu_k} V_{12}(\mu_j) = -4\alpha_1 u \left( \frac{d_{i0} + d_{i1}\mu_1}{\mu_1 - \mu_2} + \frac{d_{i0} + d_{i1}\mu_2}{\mu_2 - \mu_1} \right) = -4\alpha_1 u \cdot d_{i1}.$$

Substituting (4.28) into (4.26) gives

$$(4.29) \quad \frac{d}{dt} (\alpha_{P_0}(D_\mu)) = \frac{2}{iu} (-4\alpha_1 u) (d_{11}, d_{21}) = 8i \alpha_1 (d_{11}, d_{21}).$$

From  $\det L = -L_{11}^2 - L_{12}L_{21}$  and (2.26) write

$$L_{11} = -i(\lambda^3 + \lambda(\alpha_1 - \frac{1}{2}u^2)), \quad L_{12} = u(\lambda^2 + a\lambda + b), \quad L_{21} = -u(\lambda^2 - a\lambda + b),$$

with  $a = i\frac{u_x}{2u}$  and  $b = -\frac{u_{xx}}{4u} + \alpha_1 - \frac{1}{2}u^2$ . Compute the leading terms in  $\lambda$ :

$$-L_{11}^2 = (\lambda^3 + \lambda(\alpha_1 - \frac{1}{2}u^2))^2 = \lambda^6 + 2(\alpha_1 - \frac{1}{2}u^2)\lambda^4 + \mathcal{O}(\lambda^2),$$

and

$$-L_{12}L_{21} = u^2(\lambda^4 + \mathcal{O}(\lambda^2)) = u^2\lambda^4 + \mathcal{O}(\lambda^2).$$

Hence

$$\det L = \lambda^6 + [2(\alpha_1 - \frac{1}{2}u^2) + u^2]\lambda^4 + \mathcal{O}(\lambda^2) = \lambda^6 + 2\alpha_1\lambda^4 + \mathcal{O}(\lambda^2).$$

On the other hand (2.28) gives

$$\det L = \prod_{j=1}^3 (\lambda^2 - \lambda_j^2) = \lambda^6 - \left( \sum_{j=1}^3 \lambda_j^2 \right) \lambda^4 + \mathcal{O}(\lambda^2).$$

Equating the  $\lambda^4$ -coefficients yields  $\alpha_1 = -\frac{1}{2} \sum_{j=1}^3 \lambda_j^2$ .

## Week 5: Algebraic-Geometric Solutions for the mKdV Equation (5)

2025-10-21

**Objectives:** Express  $\omega_2$ ,  $\omega_3$ , and  $\hat{U}$  via the Riemann bilinear relations, and obtain the identities relating  $\alpha_{P_0}(D_{\mu^*})$  and  $\alpha_{P_0}(D_{\mu})$  from Abel's theorem.

**Riemann bilinear relation.**

(i) Let  $\hat{\Omega}$  and  $\tilde{\Omega}$  be (multi-valued) Abelian integrals on the Riemann surface  $R_2$ , and write their corresponding differentials as  $d\hat{\Omega}$  and  $d\tilde{\Omega}$ . For the standard canonical homology basis  $\{a_i, b_i\}_{i=1}^2$  the following identities hold by elementary contour rearrangements:

$$(5.1) \quad \oint_{a_i} \hat{\Omega} d\tilde{\Omega} + \oint_{a_i^{-1}} \hat{\Omega} d\tilde{\Omega} = \oint_{a_i} \left( \int_{z_0}^z d\hat{\Omega} \right) d\tilde{\Omega} + \oint_{a_i^{-1}} \left( \int_{z_0}^{z'} d\hat{\Omega} \right) d\tilde{\Omega}$$

$$(5.2) \quad = - \oint_{a_i} \left( \int_z^{z'} d\hat{\Omega} \right) d\tilde{\Omega} = - \left( \oint_{a_i} d\hat{\Omega} \right) \left( \oint_{b_i} d\tilde{\Omega} \right),$$

$$(5.3) \quad \oint_{b_i} \hat{\Omega} d\tilde{\Omega} + \oint_{b_i^{-1}} \hat{\Omega} d\tilde{\Omega} = \oint_{b_i} \left( \int_{z'}^z d\hat{\Omega} \right) d\tilde{\Omega} = \left( \oint_{a_i} d\hat{\Omega} \right) \left( \oint_{b_i} d\tilde{\Omega} \right).$$

Applying Stokes' theorem to the boundary of the canonical polygon for  $R_2$  gives

$$(5.4) \quad 0 = \int_{R_2} d(\hat{\Omega} d\tilde{\Omega}) = \int_{\partial R_2} \hat{\Omega} d\tilde{\Omega} = \sum_{i=1}^2 \left( \oint_{a_i} + \oint_{b_i} + \oint_{a_i^{-1}} + \oint_{b_i^{-1}} \right) \hat{\Omega} d\tilde{\Omega},$$

and hence the usual bilinear identity

$$(5.5) \quad \sum_{i=1}^2 \left( \oint_{a_i} d\hat{\Omega} \oint_{b_i} d\tilde{\Omega} - \oint_{b_i} d\hat{\Omega} \oint_{a_i} d\tilde{\Omega} \right) = 0.$$

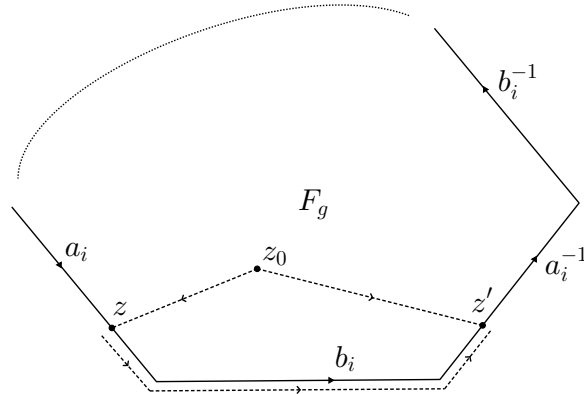


FIGURE 2. To the proof of the Riemann bilinear relations.

(ii) If the product  $\hat{\Omega} d\tilde{\Omega}$  has poles (only) at the points  $\infty^\pm$ , then performing the same computation on  $R_2 \setminus \{\infty^\pm\}$  introduces the boundary integral around small loops encircling the poles. One obtains

$$(5.6) \quad \sum_{i=1}^2 \left( \oint_{a_i} d\hat{\Omega} \oint_{b_i} d\tilde{\Omega} - \oint_{b_i} d\hat{\Omega} \oint_{a_i} d\tilde{\Omega} \right) + \oint_{l+\pi(l)} \hat{\Omega} d\tilde{\Omega} = 0,$$

where  $l + \pi(l)$  denotes the small contours around the poles at  $\infty^\pm$  with the orientation induced by the boundary of  $R_2 \setminus \{\infty^\pm\}$ .

**Applications.** Recall the Abelian integrals  $\Omega_2, \Omega_3$  from (4.8)–(4.9). Define

$$\hat{U} = (U_1, U_2), \quad U_i = \oint_{b_i} d\Omega_2, \quad \hat{V} = (V_1, V_2), \quad V_i = \oint_{b_i} d\Omega_3.$$

(1) The constants  $\omega_2, \omega_3$ . Choose

$$\hat{\Omega} = \Omega_2, \quad d\tilde{\Omega} = \frac{\lambda^2}{y} d\lambda.$$

Since  $\oint_{a_i} d\Omega_2 = 0$  and  $\oint_{b_i} d\Omega_2 = U_i$ , the bilinear identity (5.6) becomes

$$(5.7) \quad - \sum_{i=1}^2 U_i \oint_{a_i} \frac{\lambda^2}{y} d\lambda + \oint_{l+\pi(l)} \Omega_2 \frac{\lambda^2}{y} d\lambda = 0.$$

The boundary integral is the sum of residues at  $\infty^\pm$ :

$$(5.8) \quad \oint_{l+\pi(l)} \Omega_2 \frac{\lambda^2}{y} d\lambda = 2\pi i \left( \text{Res}_{\lambda=\infty^+} \Omega_2 \frac{\lambda^2}{y} d\lambda + \text{Res}_{\lambda=\infty^-} \Omega_2 \frac{\lambda^2}{y} d\lambda \right).$$

Using (4.8) and the large- $\lambda$  expansion of  $y$ , one obtains

$$(5.9) \quad \Omega_2 \frac{\lambda^2}{y} = (\lambda + \omega_2 + o(1)) \lambda^{-1} \left( 1 + \frac{1}{2}(\lambda_1^2 + \lambda_2^2 + \lambda_3^2) \lambda^{-2} + O(\lambda^{-4}) \right) = 1 + \frac{\omega_2}{\lambda} + \dots$$

as  $P \rightarrow \infty^\pm$ . Thus the residue at each of  $\infty^\pm$  equals  $\omega_2$ .

Substitution of (5.8) and (5.9) into (5.7) yields

$$(5.10) \quad \omega_2 = -\frac{i}{4\pi} \sum_{j=1}^2 U_j \oint_{a_j} \frac{\lambda^2}{y} d\lambda.$$

Repeating the computation with  $\hat{\Omega} = \Omega_3$  gives

$$(5.11) \quad \omega_3 = -\frac{i}{4\pi} \sum_{j=1}^2 V_j \oint_{a_j} \frac{\lambda^2}{y} d\lambda.$$

(2) The vector  $\hat{U}$ . We now verify that

$$\hat{U} = (U_1, U_2) = (2d_{11}, 2d_{21}), \quad U_1 = 2d_{11}.$$

Choose  $\hat{\Omega} = \Omega_2$  and

$$\tilde{\Omega} = w_1 d\lambda = (d_{10} + d_{11}\lambda) y^{-1} d\lambda, \quad \oint_{a_j} w_i d\lambda = 2\pi i \delta_{ij}.$$

Applying the bilinear identity gives

$$\sum_{i=1}^2 \left( \oint_{a_i} d\Omega_2 \oint_{b_i} w_1 d\lambda - \oint_{b_i} d\Omega_2 \oint_{a_i} w_1 d\lambda \right) + 2\pi i (\text{Res}_{\infty^+} \Omega_2 w_1 d\lambda + \text{Res}_{\infty^-} \Omega_2 w_1 d\lambda) = 0.$$

To determine the residues, use the large- $\lambda$  expansions of  $\Omega_2$  and  $y$ :

$$\Omega_2 w_1 = (\lambda + \omega_2 + o(1))(d_{10} + d_{11}\lambda) \lambda^{-3} \left( 1 + \frac{1}{2}(\lambda_1^2 + \lambda_2^2 + \lambda_3^2) \lambda^{-2} + O(\lambda^{-4}) \right) = \frac{d_{11}}{\lambda} + \dots$$

as  $P \rightarrow \infty^\pm$ . Hence each of the points  $\infty^\pm$  contributes a residue  $d_{11}$ , and the total equals  $2d_{11}$ . Since

$$\oint_{a_i} d\Omega_2 = 0, \quad \oint_{b_i} d\Omega_2 = U_i, \quad \oint_{a_i} w_1 d\lambda = 2\pi i \delta_{1i},$$

the bilinear identity reduces to

$$-2\pi i U_1 + 2\pi i (2d_{11}) = 0, \quad \text{so} \quad U_1 = 2d_{11}.$$

The same computation with  $w_2$  gives  $U_2 = 2d_{21}$ .

**Abel's theorem.** Let  $M(R_2)$  denote the meromorphic functions on  $R_2$ , and  $M'(R_2)$  the meromorphic 1-forms. For  $f \in M(R_2) \setminus \{0\}$  and  $P \in R_2$ , the order  $\text{ord}_P(f)$  is the integer  $k$  such that, in a local coordinate  $z$  with  $z(P) = 0$ ,

$$f(z) = z^k g(z), \quad g(0) \neq 0.$$

The divisor of  $f$  is

$$(f) = \sum_{P \in R_2} \text{ord}_P(f) P,$$

which is a finite sum because  $f$  has only finitely many zeros and poles. The divisor  $(w)$  of a meromorphic 1-form  $w \in M'(R_2) \setminus \{0\}$  is defined analogously by using its local expression  $w(z) = z^k h(z) dz$  with  $h(0) \neq 0$ .

A meromorphic function is holomorphic if and only if  $(f) \geq 0$ . A meromorphic 1-form is holomorphic (of the first kind) if and only if  $(w) \geq 0$ . A divisor of the form  $(f)$  is called principal; a divisor of the form  $(w)$  is canonical.

**Theorem 1** (Abel's theorem). *A divisor  $D$  is principal if and only if  $\deg D = 0$  and  $\alpha_{P_0}(D) = 0$ .*

As an example,

$$\text{Div}(r_1(x, t; P)) = D_{\infty^-, \mu^*} - D_{\infty^+, \mu},$$

which has degree zero. Abel's theorem then gives

$$\alpha_{P_0}(D_{\mu^*}) + A_{P_0}(\infty^-) = \alpha_{P_0}(D_{\infty^-, \mu^*}) = \alpha_{P_0}(D_{\infty^+, \mu}) = \alpha_{P_0}(D_\mu) + A_{P_0}(\infty^+),$$

hence

$$\alpha_{P_0}(D_{\mu^*}) = \alpha_{P_0}(D_\mu) + A_{\infty^-}(\infty^+),$$

where

$$A_{P_0}(P) = \left( \int_{P_0}^P w_1 d\lambda, \int_{P_0}^P w_2 d\lambda \right).$$

## Week 6: Algebraic-Geometric Solutions for the mKdV Equation (6)

2025-10-28

**Objectives:** Focus on Case 2. Determine the homology basis under the involution  $\tau$ , and compute the conjugate quantities  $\overline{\ln(\omega_1)}$ ,  $\overline{\omega_2}$ ,  $\overline{\omega_3}$ , and  $\overline{B_{ij}}$ .

Recall the definition (3.3):

$$(6.1) \quad R_2 := \left\{ (\lambda, y) \mid y^2 = (\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2) \right\}, \quad P = (\lambda, y).$$

We work in **Case 2** (one nonreal quadruple and one purely imaginary pair). Introduce the involutions

$$(6.2) \quad \tau_1 : (\lambda, y) \mapsto (\lambda, -y), \quad \tau : (\lambda, y) \mapsto (\bar{\lambda}, \bar{y}),$$

where  $\tau_1$  is the hyperelliptic (sheet-exchange) involution and  $\tau$  is the antiholomorphic involution; both satisfy

$$\tau_1^2 = \tau^2 = \mathbb{I}.$$

Fix a canonical homology basis  $\{a_1, a_2; b_1, b_2\}$  on  $R_2$  with the intersection properties

$$(6.3) \quad a_i \circ a_j = 0, \quad b_i \circ b_j = 0, \quad a_i \circ b_j = \delta_{ij}, \quad i, j = 1, 2.$$

The antiholomorphic involution  $\tau$  acts by sign change on the  $a$ -cycles and acts on the  $b$ -cycles by a combination of  $a$ - and  $b$ -cycles. Write

$$(6.4) \quad \tau(a_i) = -a_i, \quad \tau(b_j) = \sum_{k=1}^2 \alpha_{jk} a_k + \sum_{k=1}^2 \beta_{jk} b_k, \quad j = 1, 2,$$

for some integers  $\alpha_{jk}, \beta_{jk}$ . Using the intersection pairing (6.3) we determine these coefficients as follows.

Compute the intersections with  $a_1, a_2$ :

$$a_1 \circ \tau(b_1) = \beta_{11}(a_1 \circ b_1) = \beta_{11}, \quad a_2 \circ \tau(b_1) = \beta_{12}(a_2 \circ b_2) = \beta_{12}.$$

But the geometry of the antiholomorphic involution in this configuration yields

$$a_1 \circ \tau(b_1) = 1, \quad a_2 \circ \tau(b_1) = 0,$$

hence

$$(6.5) \quad \beta_{11} = 1, \quad \beta_{12} = 0.$$

Similarly, intersecting  $\tau(b_1)$  with the  $b$ -cycles gives

$$\tau(b_1) \circ b_1 = \alpha_{11} = 0, \quad \tau(b_1) \circ b_2 = \alpha_{12} = 1,$$

so that  $\alpha_{11} = 0, \alpha_{12} = 1$ . Collecting these values yields

$$(6.6) \quad \tau(b_1) = b_1 + a_2.$$

By the same type of computation one obtains

$$(6.7) \quad \tau(b_2) = b_2 + a_1.$$

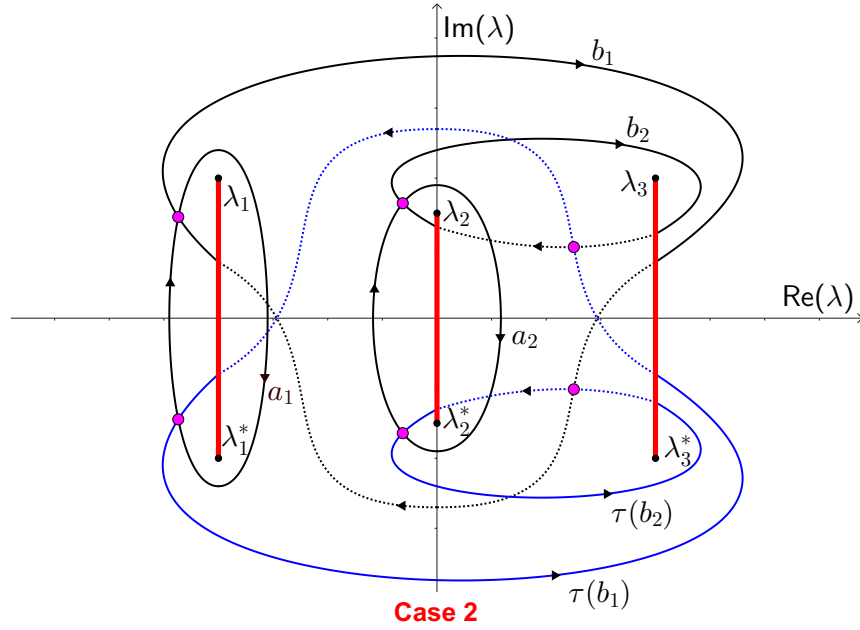


FIGURE 3. To the computations of  $\tau(b_1), \tau(b_2)$  for Case 2.

Denote by  $P_{\pm}$  the two points above the same  $\lambda$ -value on the two sheets of  $R_2$ . Let  $C_P$  be the oriented path which starts at  $P_-$  (on the second sheet), crosses the cut joining the branch points  $\lambda_3, \lambda_3^*$  through their midpoint, then continues on the first sheet and winds counterclockwise along the cut until it reaches  $P_+$ . (This is the path used below to connect  $\infty^-$  to  $\infty^+$  in the indicated way.)

The path  $C_P$  is transformed by  $\tau$  as

$$(6.8) \quad \tau(C_P) = C_P + a_3,$$



since  $\tau(C_P) - C_P = a_3$ . Computing the intersections  $a_3 \circ b_j$  from the geometry one finds

$$a_3 \circ b_1 = -1, \quad a_3 \circ b_2 = -1,$$

and therefore

$$(6.9) \quad a_3 = -a_1 - a_2.$$

In fact one must include the small loop around  $\infty^+$ . Thus the correct homology relation is

$$(6.10) \quad a_3 = -a_1 - a_2 + \ell,$$

where  $\ell$  denotes a small positively oriented loop encircling  $\infty^+$ .

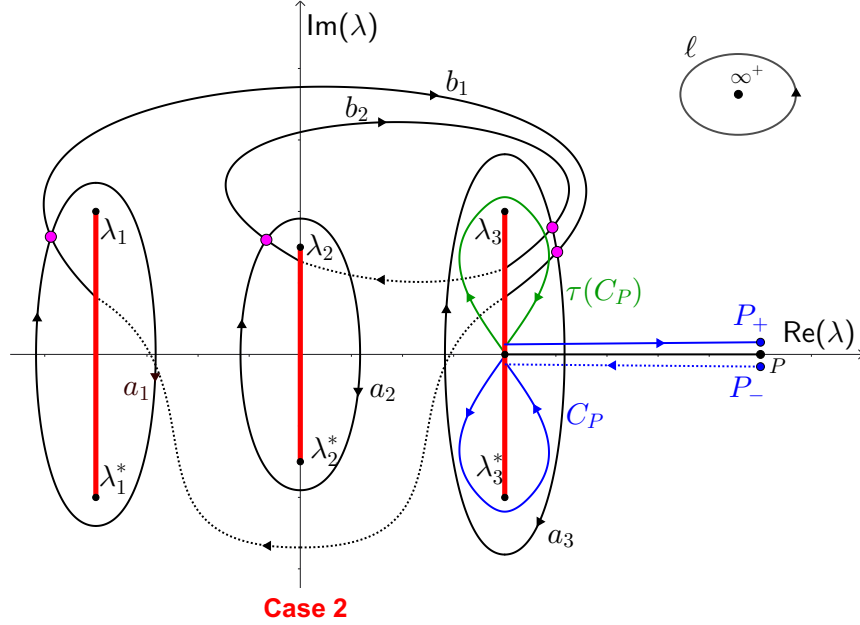


FIGURE 4. To the computation of  $\tau(C_P)$  for Case 2.

Let  $d\Omega_1$  be the normalized (third-kind) Abelian differential whose  $a$ -periods vanish:

$$(6.11) \quad \oint_{a_i} d\Omega_1 = 0, \quad i = 1, 2,$$

and which has the prescribed logarithmic behaviour at the points at infinity:

$$(6.12) \quad \Omega_1(P) = \int_{P_0}^P d\Omega_1 = \pm (\ln \lambda + \ln \omega_1 + o(1)), \quad P \rightarrow \infty^\pm,$$

where  $\omega_1$  is the (nonzero) constant appearing in the local expansion and  $w_i$  will denote holomorphic differentials as before.

Since  $d\Omega_1$  is normalized by  $\oint_{a_i} d\Omega_1 = 0$  and has residue 1 at  $\infty^+$ , we obtain

$$(6.13) \quad \oint_{\ell} d\Omega_1 = 2\pi i.$$

Therefore

$$(6.14) \quad \oint_{a_3} d\Omega_1 = \oint_{-a_1-a_2+\ell} d\Omega_1 = -\sum_{i=1}^2 \oint_{a_i} d\Omega_1 + \oint_{\ell} d\Omega_1 = 0 + 2\pi i = 2\pi i.$$

Using the path  $C_P$  joining  $\infty^-$  to  $\infty^+$  and the relation  $\tau(C_P) = C_P + a_3$ , the asymptotics (6.12) imply that

$$(6.15) \quad \lim_{P \rightarrow \infty^+} \left( \int_{\infty^-}^{\infty^+} d\Omega_1 - 2 \ln \lambda \right) = \lim_{P \rightarrow \infty^+} \left( \int_{C_P} d\Omega_1 - 2 \ln \lambda \right) = 2 \ln \omega_1.$$

We compute the conjugate quantity

$$(6.16) \quad \begin{aligned} 2 \overline{\ln(\omega_1)} &= \lim_{P \rightarrow \infty^+} \left( \int_{C_P} \overline{d\Omega_1(P)} - 2 \ln \lambda \right) = \lim_{P \rightarrow \infty^+} \left( \int_{\tau(C_P)} d\Omega_1 - 2 \ln \lambda \right) \\ &= \lim_{P \rightarrow \infty^+} \left( \int_{C_P + a_3} d\Omega_1 - 2 \ln \lambda \right) = \lim_{P \rightarrow \infty^+} \left( \int_{C_P} d\Omega_1 - 2 \ln \lambda \right) + \oint_{a_3} d\Omega_1 \\ &= 2 \ln(\omega_1) + 2\pi i, \end{aligned}$$

where in the second equality we used  $\overline{d\Omega_1(P)} = \tau^* d\Omega_1$  and  $\tau(C_P) = C_P + a_3$ , and in the last step we used (6.14). From (6.16) we obtain

$$(6.17) \quad \overline{\ln(\omega_1)} = \ln(\omega_1) + \pi i, \quad \Rightarrow \quad \Im(\ln(\omega_1)) = -\frac{\pi}{2}.$$

Hence, writing  $\omega_1$  in modulus–argument form gives

$$(6.18) \quad \ln(\omega_1) = \ln|\omega_1| - i\beta, \quad \beta > 0,$$

(and in our normalization one finds  $\beta = \frac{\pi}{2}$ ).

The other normalized Abelian integrals have the asymptotics

$$(6.19) \quad \Omega_2(P) = \pm(\lambda + \omega_2 + o(1)), \quad \Omega_3(P) = \pm(4\lambda^3 + \omega_3 + o(1)), \quad P \rightarrow \infty^\pm,$$

so that, using the same connecting path  $C_P$ ,

$$(6.20) \quad 2\omega_2 = \lim_{P \rightarrow \infty^+} \left( \int_{C_P} d\Omega_2 - 2\lambda \right), \quad 2\omega_3 = \lim_{P \rightarrow \infty^+} \left( \int_{C_P} d\Omega_3 - 8\lambda^3 \right).$$

Taking complex conjugates and using the action of  $\tau$  on  $C_P$  one deduces

$$(6.21) \quad 2\overline{\omega_2} = 2\omega_2, \quad 2\overline{\omega_3} = 2\omega_3, \quad \Rightarrow \quad \omega_2, \omega_3 \in \mathbb{R}.$$

Recall the period matrix entries are defined by

$$(6.22) \quad B_{ij} = \oint_{b_j} w_i d\lambda,$$

where  $\{w_1, w_2\}$  denote the normalized holomorphic differentials with

$$\oint_{a_k} w_i d\lambda = 2\pi i \delta_{ik}, \quad i, k = 1, 2.$$

Using the antiholomorphic involution  $\tau$  and the relations  $\tau(b_1) = b_1 + a_2$ ,  $\tau(b_2) = b_2 + a_1$  we get

$$(6.23) \quad \begin{aligned} \overline{B_{ij}} &= \oint_{b_j} \overline{w_i d\lambda} = \oint_{\tau(b_j)} w_i d\lambda = \oint_{b_j} w_i d\lambda + \sum_{l \neq j} \oint_{a_l} w_i d\lambda \\ &= B_{ij} + 2\pi i \sum_{l \neq j} \delta_{il}. \end{aligned}$$

Therefore, in matrix form,

$$(6.24) \quad \overline{B} = B + 2\pi i \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The Riemann theta function is defined by

$$(6.25) \quad \Theta(z | B) = \sum_{n \in \mathbb{Z}^2} \exp\left(\frac{1}{2} n^\top B n + n^\top z\right).$$

Using (6.24) one checks the conjugation property

$$(6.26) \quad \overline{\Theta(z \mid B)} = \Theta(\bar{z} \mid \bar{B}) = \sum_{n \in \mathbb{Z}^2} \exp\left(\frac{1}{2} n^\top B n + n^\top \bar{z} + 2\pi i n_1 n_2\right) = \Theta(\bar{z} \mid B),$$

since  $\exp(2\pi i n_1 n_2) = 1$  for integers  $n_1, n_2$ .

**Notational conventions and auxiliary differentials.** For later calculations we introduce the following integrals and differentials:

$$(6.27) \quad \mathcal{Y}_{a_i}^{[j]} = \oint_{a_i} \frac{\lambda^j}{y} d\lambda, \quad \mathcal{Y}_{b_i}^{[j]} = \oint_{b_i} \frac{\lambda^j}{y} d\lambda, \quad i = 1, 2, \quad j \geq 0,$$

$$(6.28) \quad \Omega_j(P) = \int_{(\lambda_3, 0)}^P d\Omega_j, \quad d\Omega_2^{(1)} = \sum_{i=0}^3 \frac{c_{2i} \lambda^i}{y(\lambda)} d\lambda, \quad d\Omega_3^{(1)} = \sum_{i=1}^5 \frac{c_{3i} \lambda^i}{y(\lambda)} d\lambda,$$

where  $P = (\lambda, y) \in R_2$ , the base point  $(\lambda_3, 0)$  is fixed, and the constants  $c_{2i}, c_{3i}$  (as well as the coefficients  $d_{ik}$  appearing in the holomorphic differentials) will be determined later on.

## Week 7: Algebro-Geometric Solutions for the mKdV Equation (7)

2025-11-04

**Objectives:** Focus on Case 1. Introduce two transformations used in the computation of  $\mathcal{Y}_{a_i}^{[j]}$  and  $\mathcal{Y}_{b_i}^{[j]}$ ; carry out the evaluation of  $\mathcal{Y}_{a_i}^{[0]}, \mathcal{Y}_{a_i}^{[1]}$ , and determine the coefficients  $d_{ij}$ .

Recall that

$$w_i(\lambda) = \frac{d_{i0} + d_{i1}\lambda}{y(\lambda)}, \quad y(\lambda) = \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}.$$

and

$$(7.1) \quad \oint_{a_j} w_i(\lambda) d\lambda = 2\pi i \delta_{ij}, \quad i, j = 1, 2.$$

Hence, writing the period integrals in matrix form, the normalization (7.1) is equivalent to

$$(7.2) \quad 2\pi i \mathbb{I}_2 = \begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} \begin{pmatrix} \oint_{a_1} \frac{d\lambda}{y} & \oint_{a_2} \frac{d\lambda}{y} \\ \oint_{a_1} \frac{\lambda d\lambda}{y} & \oint_{a_2} \frac{\lambda d\lambda}{y} \end{pmatrix}.$$

In particular,

$$(7.3) \quad \begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} = 2\pi i \begin{pmatrix} \oint_{a_1} \frac{d\lambda}{y} & \oint_{a_2} \frac{d\lambda}{y} \\ \oint_{a_1} \frac{\lambda d\lambda}{y} & \oint_{a_2} \frac{\lambda d\lambda}{y} \end{pmatrix}^{-1}.$$

Recall the intersection and period conventions for the canonical homology basis  $\{a_i, b_i\}_{i=1}^2$ :

$$(7.4) \quad a_i \circ a_j = b_i \circ b_j = 0, \quad a_i \circ b_j = \delta_{ij},$$

$$(7.5) \quad \oint_{b_j} w_i(\lambda) d\lambda = B_{ij}.$$

**Case 1.**  $\lambda_1, \lambda_2, \lambda_3 \in i\mathbb{R}$ .

Let  $\tau$  denote complex conjugation on the curve,

$$\tau: (\lambda, y) \mapsto (\lambda^*, y^*).$$

The action of  $\tau$  on the homology basis may be chosen (up to orientation) as

$$(7.6) \quad \tau(a_1) = -a_2, \quad \tau(a_2) = -a_1, \quad \tau(b_1) = b_2, \quad \tau(b_2) = b_1.$$

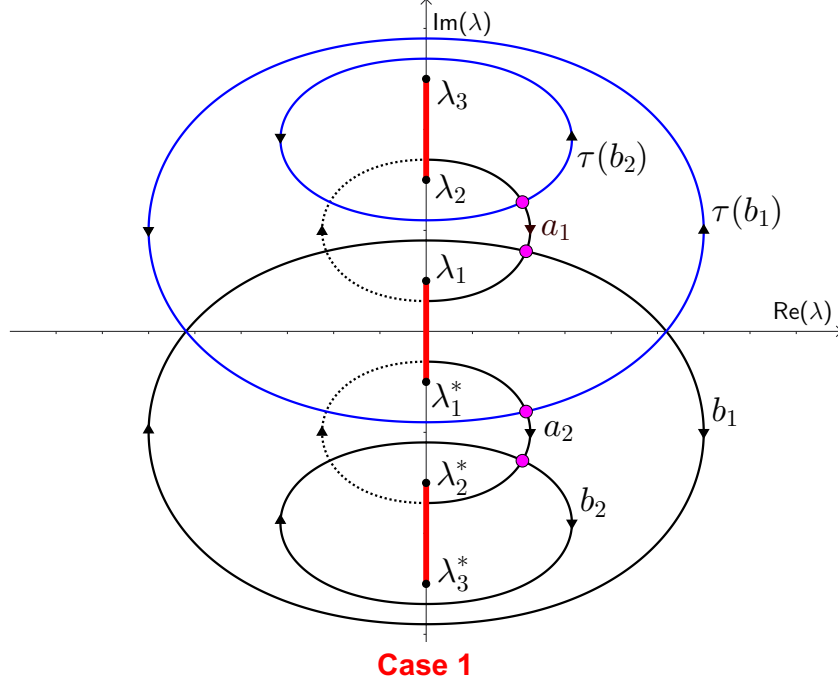


FIGURE 5. To the computations of  $\tau(b_1), \tau(b_2)$  for Case 1.

With the choice of cycles described above, the  $a$ -periods reduce to integrals along paths on the first sheet:

$$(7.7) \quad \oint_{a_1} \frac{d\lambda}{y} = 2 \int_{\lambda_2}^{\lambda_1} \frac{d\lambda}{y}, \quad \oint_{a_1} \frac{\lambda d\lambda}{y} = 2 \int_{\lambda_2}^{\lambda_1} \frac{\lambda d\lambda}{y},$$

and

$$(7.8) \quad \oint_{a_2} \frac{d\lambda}{y} = 2 \int_{\lambda_1^*}^{\lambda_2^*} \frac{d\lambda}{y}, \quad \oint_{a_2} \frac{\lambda d\lambda}{y} = 2 \int_{\lambda_1^*}^{\lambda_2^*} \frac{\lambda d\lambda}{y}.$$

Recall the definition

$$(7.9) \quad y(\lambda) = \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}, \quad \Lambda = \lambda^2, \quad d\Lambda = 2\lambda d\lambda.$$

Applying the substitution  $\Lambda = \lambda^2$  transforms the basic integral into a rational function of  $\Lambda$ :

$$(7.10) \quad \int \frac{d\lambda}{y(\lambda)} = \int \frac{d\lambda}{\sqrt{\prod_{i=1}^3 (\lambda^2 - \lambda_i^2)}} = \int \frac{d\Lambda}{2\sqrt{\Lambda \prod_{i=1}^3 (\Lambda - \lambda_i^2)}}.$$

Thus the problem reduces to integrals of the general form

$$(7.11) \quad \int \frac{d\Lambda}{\sqrt{\prod_{i=1}^4 (\Lambda - \beta_i)}}, \quad \beta_1 < \beta_2 < \beta_3 < \beta_4.$$

To reduce (7.11) to the Legendre normal form we use the change of variable

$$(7.12) \quad z^2 = \frac{\beta_3 - \beta_1}{\beta_4 - \beta_1} \frac{\Lambda - \beta_4}{\Lambda - \beta_3}.$$

A direct computation yields the standard reduction

$$(7.13) \quad \int \frac{d\Lambda}{\sqrt{\prod_{i=1}^4 (\Lambda - \beta_i)}} = \frac{2}{\sqrt{(\beta_3 - \beta_1)(\beta_4 - \beta_2)}} \int \frac{dz}{\sqrt{(1 - z^2)(1 - k^2 z^2)}},$$

with the elliptic modulus given by

$$(7.14) \quad k^2 = \frac{(\beta_3 - \beta_2)(\beta_4 - \beta_1)}{(\beta_3 - \beta_1)(\beta_4 - \beta_2)}.$$

For reference, the incomplete elliptic integrals in Jacobi form are

$$F(z, k) = \int_0^z \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}}, \quad E(z, k) = \int_0^z \sqrt{\frac{1-k^2t^2}{1-t^2}} dt, \quad 0 < k < 1,$$

and the elliptic integral of the third kind is

$$\Pi(z, \alpha^2, k) = \int_0^z \frac{dt}{(1-\alpha^2t^2)\sqrt{(1-t^2)(1-k^2t^2)}}.$$

These standard forms allow the evaluation of the  $a$ -period integrals (7.7)–(7.8) once the correspondence between the branch points  $\{\beta_i\}$  and the original  $\lambda$ -branch points is fixed.

Applying (7.13)–(7.14) to our concrete problem: we may take

$$\beta_1 = \lambda_3^2, \quad \beta_2 = \lambda_2^2, \quad \beta_3 = \lambda_1^2, \quad \beta_4 = 0,$$

and (using (7.12)) obtain

$$(7.15) \quad z^2 = \frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2} \frac{\Lambda}{\Lambda - \lambda_1^2} = \frac{(\lambda_1^2 - \lambda_3^2) \lambda^2}{\lambda_3^2 (\lambda_1^2 - \lambda^2)}.$$

For these choices the modulus becomes

$$(7.16) \quad k_1^{(1)2} = \frac{(\lambda_1^2 - \lambda_2^2)(0 - \lambda_3^2)}{(\lambda_1^2 - \lambda_3^2)(0 - \lambda_2^2)} = \frac{\lambda_3^2}{\lambda_2^2} \frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2},$$

so that

$$(7.17) \quad \int \frac{d\lambda}{y(\lambda)} = \frac{1}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int \frac{dz}{\sqrt{(1-z^2)(1-k_1^{(1)2}z^2)}}, \quad k_1^{(1)} = \frac{\lambda_3}{\lambda_2} \sqrt{\frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2}}$$

Defining the incomplete elliptic integral

$$(7.18) \quad v = F(z, k_1^{(1)}) = \int_0^z \frac{dt}{\sqrt{(1-t^2)(1-k_1^{(1)2}t^2)}}, \quad z = \operatorname{sn}(v, k_1^{(1)}),$$

we obtain the standard elliptic parametrisation of  $\lambda$  via (7.15) and (7.18).

Recall the relations:

$$\operatorname{sn}^2(u, k) + \operatorname{cn}^2(u, k) = 1, \quad \operatorname{dn}^2(u, k) + k^2 \operatorname{sn}^2(u, k) = 1,$$

and the complete elliptic integrals

$$K(k) = F(1, k), \quad K'(k) = F(1, k'), \quad k'^2 = 1 - k^2.$$

From (7.15) one checks the mapping of special  $\Lambda$ -values to  $z$ -values:

$$\Lambda = \lambda^2 = 0 \leftrightarrow z = 0 \leftrightarrow v = 0, \quad \Lambda = \lambda_1^2 \leftrightarrow z^2 \rightarrow \infty \leftrightarrow v = \pm iK_1^{(1)'}$$

and

$$\Lambda = \lambda^2 = \lambda_2^2 \leftrightarrow z^2 = \frac{\lambda_2^2}{\lambda_3^2} \frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda_2^2} = \frac{1}{k_1^{(1)2}} \leftrightarrow v = \pm (K_1^{(1)} + iK_1^{(1)'}).$$

With this choice, the  $a_1$ -period can be expressed in the  $v$ -variable as an integral between two points in the complex  $v$ -plane. In particular, we have

$$(7.19) \quad 2 \int_{\lambda_2}^{\lambda_1} \frac{d\lambda}{y(\lambda)} = \frac{2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_{K_1^{(1)} + iK_1^{(1)'}}^{iK_1^{(1)'}} dv = \frac{-2K_1^{(1)}}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} = \frac{-2iK_1^{(1)'}}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \in \mathbb{R}^-.$$

The symmetry under complex conjugation,  $\tau : (\lambda, y) \mapsto (\lambda^*, y^*)$ , implies the relations

$$(7.20) \quad \left( \oint_{a_2} \frac{d\lambda}{y} \right)^* = \oint_{a_2} \tau^* \frac{d\lambda}{y} = \oint_{\tau(a_2)} \frac{d\lambda}{y} = \oint_{-a_1} \frac{d\lambda}{y} = - \oint_{a_1} \frac{d\lambda}{y} \in \mathbb{R}^+.$$

Similarly, one finds

$$(7.21) \quad \oint_{a_2} \frac{\lambda d\lambda}{y} = \oint_{a_1} \frac{\lambda d\lambda}{y}.$$

Using the substitution  $\Lambda = \lambda^2$ , one rewrites

$$(7.22) \quad \int \frac{\lambda d\lambda}{y(\lambda)} = \int \frac{d\Lambda}{2\sqrt{\prod_{i=1}^3 (\Lambda - \lambda_i^2)}}.$$

Introducing the change of variables

$$(7.23) \quad z^2 = \frac{\beta_1 - \beta_3}{\Lambda - \beta_3},$$

the integral becomes

$$(7.24) \quad \int \frac{\lambda d\lambda}{y(\lambda)} = \frac{1}{-i\sqrt{\lambda_1^2 - \lambda_3^2}} \int \frac{dz}{\sqrt{(1-z^2)(1-k_2^{(1)2}z^2)}}, \quad k_2^{(1)} = \sqrt{\frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2}}.$$

Hence,

$$(7.25) \quad \oint_{a_1} \frac{\lambda d\lambda}{y} = \frac{-2i K_2^{(1)}}{\sqrt{\lambda_1^2 - \lambda_3^2}} = \oint_{a_2} \frac{\lambda d\lambda}{y}.$$

Since

$$(7.26) \quad \begin{pmatrix} \oint_{a_1} \frac{d\lambda}{y} & \oint_{a_2} \frac{d\lambda}{y} \\ \oint_{a_1} \frac{\lambda d\lambda}{y} & \oint_{a_2} \frac{\lambda d\lambda}{y} \end{pmatrix} = \begin{pmatrix} \oint_{a_1} \frac{d\lambda}{y} & 0 \\ 0 & \oint_{a_1} \frac{\lambda d\lambda}{y} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix},$$

the relation (7.3) becomes

$$\begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} = \pi i \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \oint_{a_1} \frac{d\lambda}{y} & 0 \\ 0 & \oint_{a_1} \frac{\lambda d\lambda}{y} \end{pmatrix}^{-1}.$$

This yields the coefficients

$$(7.27) \quad \begin{aligned} d_{10} &= -\frac{\pi \lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}}{2 K_1^{(1)}}, & d_{11} &= -\frac{\pi \sqrt{\lambda_1^2 - \lambda_3^2}}{2 K_2^{(1)}}, \\ d_{20} &= -d_{10}, & d_{21} &= d_{11}. \end{aligned}$$

### Supplementary computations:

Define

$$(7.28) \quad F(z, k) = \int_0^z \frac{dt}{f(t, k)}, \quad f(t, k) = \sqrt{(1-t^2)(1-k^2t^2)}, \quad 0 < k < 1,$$

where the principal branch of the square root is chosen and we fix  $f(0, k) = 1$ .

Set

$$(7.29) \quad K(k) = F(1, k), \quad K'(k) = K(k'), \quad k' = \sqrt{1-k^2}.$$

We take the segments  $[-k^{-1}, -1] \cup [1, k^{-1}]$  as branch cuts for  $f(t, k)$  on the extended real axis. Starting at  $t = 0$  and moving right along the real axis, each time one crosses a branch point an extra phase factor  $e^{-\pi i/2} = -i$  is accumulated; moving left from 0 produces the conjugate factor  $e^{\pi i/2} = i$ . With these conventions,  $f(t, k)$  on the first sheet admits the explicit real-axis description

$$(7.30) \quad f(t, k) = \begin{cases} \sqrt{(1-t^2)(1-k^2t^2)} \in \mathbb{R}^+, & t \in (-1, 1), \\ -i\sqrt{(t^2-1)(1-k^2t^2)} \in i\mathbb{R}^-, & t \in (1, k^{-1}), \\ i\sqrt{(t^2-1)(1-k^2t^2)} \in i\mathbb{R}^+, & t \in (-k^{-1}, -1), \\ -\sqrt{(t^2-1)(k^2t^2-1)} \in \mathbb{R}^-, & t \in (-\infty, -k^{-1}) \cup (k^{-1}, \infty). \end{cases}$$

We now evaluate several definite integrals that will be used below.

For the interval  $(k^{-1}, \infty)$  perform the substitution

$$(7.31) \quad t = k^{-1}s^{-1}, \quad dt = -k^{-1}s^{-2}ds,$$

which maps  $t \in [k^{-1}, \infty)$  to  $s \in (0, 1]$ . Direct computation gives

$$(7.32) \quad \begin{aligned} \int_{k^{-1}}^{\infty} \frac{dt}{\sqrt{(t^2-1)(k^2t^2-1)}} &= \int_1^0 \frac{-k^{-1}s^{-2}ds}{\sqrt{(k^{-2}s^{-2}-1)(s^{-2}-1)}} \\ &= \int_0^1 \frac{ds}{\sqrt{(1-s^2)(1-k^2s^2)}} \\ &= F(1, k) = K(k). \end{aligned}$$

For the finite interval  $(1, k^{-1})$  use the substitution

$$(7.33) \quad t = (1 - (k')^2s^2)^{-1/2}, \quad dt = (k')^2s(1 - (k')^2s^2)^{-3/2}ds,$$

which sends  $s \in [0, 1]$  to  $t \in [1, k^{-1}]$ . Under this change of variables

$$(7.34) \quad \begin{aligned} \sqrt{(t^2-1)(1-k^2t^2)} &= \sqrt{\frac{(k's)^2}{1-(k's)^2} \cdot (k')^2 \frac{1-s^2}{1-(k's)^2}} \\ &= (k')^2s \frac{\sqrt{1-s^2}}{1-(k's)^2}. \end{aligned}$$

Hence

$$(7.35) \quad \begin{aligned} \int_1^{k^{-1}} \frac{dt}{\sqrt{(t^2-1)(1-k^2t^2)}} &= \int_0^1 \frac{ds}{\sqrt{(1-s^2)(1-(k')^2s^2)}} \\ &= F(1, k') = K(k') = K'(k). \end{aligned}$$

Combining the evaluations on  $[0, 1]$  and  $(1, k^{-1})$ , and recalling the phase  $-i$  on  $(1, k^{-1})$  from (7.30), we obtain

$$(7.36) \quad \begin{aligned} F(k^{-1}, k) &= \int_0^1 \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}} + i \int_1^{k^{-1}} \frac{dt}{\sqrt{(t^2-1)(1-k^2t^2)}} \\ &= K(k) + iK'(k). \end{aligned}$$

Taking the limit  $z \rightarrow +\infty$  along the positive real axis (passing through  $k^{-1}$ ) yields

$$(7.37) \quad F(+\infty, k) = K(k) + iK'(k) - K(k) = iK'(k),$$

while symmetry across the origin gives

$$(7.38) \quad F(-1, k) = -F(1, k) = -K(k).$$

Similarly, evaluating  $F$  at  $-k^{-1}$  (follow the path from 0 to  $-1$  to  $-k^{-1}$  and use the phase convention on  $(-k^{-1}, -1)$  in (7.30)) yields

$$(7.39) \quad \begin{aligned} F(-k^{-1}, k) &= -K(k) + \int_{-1}^{-k^{-1}} \frac{dt}{i\sqrt{(t^2-1)(1-k^2t^2)}} \\ &= -K(k) + i \int_1^{k^{-1}} \frac{dt}{\sqrt{(t^2-1)(1-k^2t^2)}} \\ &= -K(k) + iK'(k). \end{aligned}$$

Finally, taking  $z \rightarrow -\infty$  along the negative real axis (passing through  $-k^{-1}$ ) gives

$$(7.40) \quad F(-\infty, k) = -K(k) + iK'(k) + K(k) = iK'(k).$$

Since

$$(type\ 1) \quad \int \frac{\lambda^{2n}}{y(\lambda)} d\lambda = \int \frac{\Lambda^n d\Lambda}{2(\Lambda(\Lambda - \lambda_1^2)(\Lambda - \lambda_2^2)(\Lambda - \lambda_3^2))^{1/2}}, \quad \Lambda = \lambda^2,$$

and

$$(type\ 2) \quad \int \frac{\lambda^{2n+1}}{y(\lambda)} d\lambda = \int \frac{\Lambda^n d\Lambda}{2((\Lambda - \lambda_1^2)(\Lambda - \lambda_2^2)(\Lambda - \lambda_3^2))^{1/2}}, \quad \Lambda = \lambda^2,$$

we treat the two types of integrals separately.

**Computation of Type 1 Integrals  $(\mathcal{Y}_{a_i}^{[0]}, \mathcal{Y}_{b_i}^{[0]})$ .** Introduce a change of variable from  $\Lambda$  to  $z$  by

$$(7.41) \quad z^2 = \frac{\beta_3 - \beta_1}{\beta_4 - \beta_1} \frac{\Lambda - \beta_4}{\Lambda - \beta_3}, \quad \beta_1 < \beta_2 < \beta_3 < \beta_4.$$

Solving (7.41) for  $\Lambda$  yields

$$(7.42) \quad \Lambda = \frac{z^2\beta_3(\beta_4 - \beta_1) - (\beta_3 - \beta_1)\beta_4}{z^2(\beta_4 - \beta_1) - (\beta_3 - \beta_1)} = \beta_3 + \frac{(\beta_3 - \beta_1)(\beta_3 - \beta_4)}{(\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1)}.$$

Differentiation gives

$$(7.43) \quad d\Lambda = \frac{2(\beta_3 - \beta_1)(\beta_4 - \beta_3)(\beta_4 - \beta_1)z}{((\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1))^2} dz.$$

The shifted factors  $\Lambda - \beta_i$  become rational functions of  $z$ . In particular

$$(7.44) \quad \Lambda - \beta_1 = \frac{(\beta_3 - \beta_1)(\beta_4 - \beta_1)(z^2 - 1)}{(\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1)},$$

$$(7.45) \quad \Lambda - \beta_2 = \frac{(\beta_3 - \beta_2)(\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1)(\beta_4 - \beta_2)}{(\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1)},$$

$$(7.46) \quad \Lambda - \beta_4 = \frac{(\beta_3 - \beta_4)(\beta_4 - \beta_1)z^2}{(\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1)}.$$



Hence the full quartic factor factors as

$$(7.47) \quad \prod_{i=1}^4 (\Lambda - \beta_i) = \frac{(\beta_3 - \beta_1)^2 (\beta_4 - \beta_1)^2 (\beta_4 - \beta_3)^2 z^2 (z^2 - 1)}{((\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1))^4} \\ \times \left( (\beta_3 - \beta_2)(\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1)(\beta_4 - \beta_2) \right).$$

Define the modulus

$$(7.48) \quad k_1^2 = \frac{(\beta_3 - \beta_2)(\beta_4 - \beta_1)}{(\beta_3 - \beta_1)(\beta_4 - \beta_2)}.$$

Taking a principal square root of (7.47) (and grouping algebraic constants) yields

$$(7.49) \quad \left( \prod_{i=1}^4 (\Lambda - \beta_i) \right)^{1/2} = \frac{(\beta_3 - \beta_1)(\beta_4 - \beta_1)(\beta_4 - \beta_3)}{((\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1))^2} \sqrt{(\beta_3 - \beta_1)(\beta_4 - \beta_2)} z \sqrt{(1 - z^2)(1 - k_1^2 z^2)}.$$

Combining (7.43) and (7.49) produces the standard elliptic reduction

$$(7.50) \quad \int \frac{d\Lambda}{\sqrt{\prod_{i=1}^4 (\Lambda - \beta_i)}} = \frac{2}{\sqrt{(\beta_4 - \beta_2)(\beta_3 - \beta_1)}} \int \frac{dz}{\sqrt{(1 - z^2)(1 - k_1^2 z^2)}},$$

valid on the appropriate choice of branches for the square roots.

Specializing the parameters by

$$(7.51) \quad \beta_1 = \lambda_3^2, \quad \beta_2 = \lambda_2^2, \quad \beta_3 = \lambda_1^2, \quad \beta_4 = 0,$$

the change of variable (7.41) becomes

$$(7.52) \quad z^2 = \frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2} \frac{\lambda^2}{\lambda^2 - \lambda_1^2}, \quad z = \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2}} \frac{\lambda}{\sqrt{\lambda^2 - \lambda_1^2}},$$

where the positive square root is taken for the principal branch. The modulus (7.48) simplifies to

$$(7.53) \quad k_1^2 = \frac{(\lambda_1^2 - \lambda_2^2)(\lambda_2^2 - \lambda_3^2)}{(\lambda_1^2 - \lambda_3^2)(0 - \lambda_2^2)} = \frac{(\lambda_1^2 - \lambda_2^2) \lambda_3^2}{(\lambda_1^2 - \lambda_3^2) \lambda_2^2}.$$

Choose the cut  $[-\lambda_1, \lambda_1]$  for the function  $\sqrt{\lambda^2 - \lambda_1^2}$  and fix the sheet by requiring

$$\sqrt{\lambda^2 - \lambda_1^2} \sim \lambda \quad (\lambda \rightarrow \infty).$$

For  $\text{Re } \lambda \geq 0$  and  $\lambda \in i\mathbb{R}$  write  $\lambda = it$  with  $t \in \mathbb{R}$ . If  $\lambda_j = it_j$  then

$$(7.54) \quad \sqrt{\lambda^2 - \lambda_1^2} = \begin{cases} i\sqrt{t^2 - t_1^2}, & t > t_1, \\ \sqrt{t_1^2 - t^2}, & -t_1 < t < t_1, \\ -i\sqrt{t^2 - t_1^2}, & t < -t_1. \end{cases}$$

Hence for  $\lambda \in i\mathbb{R}$  on the imaginary axis with  $\text{Re } \lambda \geq 0$ ,

$$(7.55) \quad z = \begin{cases} \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2}} \frac{\lambda}{\sqrt{\lambda^2 - \lambda_1^2}} \in i\mathbb{R}, & \lambda \in (-\lambda_1, \lambda_1), \\ \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2}} \frac{|\lambda|}{\sqrt{\lambda_1^2 - \lambda^2}} \in \mathbb{R}^+, & \lambda \in (-i\infty, -\lambda_1) \cup (\lambda_1, +i\infty). \end{cases}$$

The oriented real/imaginary segments map under  $\lambda \mapsto z$  as follows (the images are indicated in the  $z$ -plane):

$$(7.56) \quad \begin{aligned} (\lambda_3, \lambda_2) &\longrightarrow (1, k_1^{-1}), & (\lambda_2, \lambda_1) &\longrightarrow (k_1^{-1}, +\infty), \\ (\lambda_1, 0) &\longrightarrow (+i\infty, 0), & (0, -\lambda_1) &\longrightarrow (0, -i\infty), \\ (-\lambda_1, -\lambda_2) &\longrightarrow (+\infty, k_1^{-1}), & (-\lambda_2, -\lambda_3) &\longrightarrow (k_1^{-1}, 1). \end{aligned}$$

Since

$$(7.57) \quad \lambda y(\lambda) = \sqrt{\prod_{i=1}^4 (\Lambda - \beta_i)} = \frac{(\beta_3 - \beta_1)(\beta_4 - \beta_1)(\beta_4 - \beta_3)}{((\beta_4 - \beta_1)z^2 - (\beta_3 - \beta_1))^2} z \sqrt{(1 - z^2)(1 - k_1^2 z^2)},$$

for our chosen map between  $\lambda$  and  $z$  we may write

$$\lambda y(\lambda) = c_0 z \sqrt{(1 - z^2)(1 - k_1^2 z^2)}, \quad c_0 > 0.$$

For  $\lambda = it \in i\mathbb{R}$ , note that  $z \in (0, 1)$  when  $\lambda \in (\lambda_3, +i\infty)$ ; in this region  $\lambda y(\lambda) \in \mathbb{R}^+$ . The branch points  $\pm\lambda_3, \pm\lambda_2$  of  $y(\lambda)$  correspond to the branch points  $1, k_1^{-1}$  of  $f(z, k_1)$ , where  $f(z, k_1) = \sqrt{(1 - z^2)(1 - k_1^2 z^2)}$  and the branch cuts are chosen as in (7.28)–(7.30).

Thus, for  $\lambda \in i\mathbb{R}$ ,

$$\lambda y(\lambda) \sim \begin{cases} z f(z, k_1), & \lambda \in (-\lambda_2, +i\infty) \cup (-\lambda_3, -i\infty), \\ -z f(z, k_1), & \lambda \in (-\lambda_2, -\lambda_3). \end{cases}$$

Hence for  $\lambda \in i\mathbb{R} \setminus [-\lambda_2, -\lambda_3]$ ,

$$(7.58) \quad \frac{d\lambda}{y(\lambda)} = \frac{1}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \frac{dz}{f(z, k_1)},$$

and on the interval  $(-\lambda_2, -\lambda_3)$ , where the sign of the square root changes,

$$(7.59) \quad \frac{d\lambda}{y(\lambda)} = \frac{1}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \frac{dz}{-f(z, k_1)}.$$

Using (7.58)–(7.59) together with the standard elliptic integrals, and noting

$$(7.60) \quad \sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)} = -i \lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2} > 0,$$

one obtains the following closed forms

$$(7.61) \quad \begin{aligned} \mathcal{Y}_{a_1}^{[0]} &= 2 \int_{\lambda_2}^{\lambda_1} \frac{d\lambda}{y(\lambda)} = \frac{2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_{k_1^{-1}}^{+\infty} \frac{dz}{f(z, k_1)} \\ &= \frac{2}{-i \lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K_1 + iK_1'}^{iK_1'} dv = \frac{-2i K_1}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \in \mathbb{R}^-, \end{aligned}$$

$$(7.62) \quad \mathcal{Y}_{a_2}^{[0]} = 2 \int_{-\lambda_1}^{-\lambda_2} \frac{d\lambda}{y(\lambda)} = \frac{2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_{+\infty}^{k_1^{-1}} \frac{dz}{f(z, k_1)} = -\mathcal{Y}_{a_1}^{[0]} = \frac{2i K_1}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \in \mathbb{R}^+,$$

$$(7.63) \quad \begin{aligned} \mathcal{Y}_{b_2}^{[0]} &= 2 \int_{-\lambda_2}^{-\lambda_3} \frac{d\lambda}{y(\lambda)} = \frac{2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_{k_1^{-1}}^1 \frac{dz}{-f(z, k_1)} \\ &= \frac{2}{-i \lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K_1}^{K_1 + iK_1'} dv = \frac{-2K_1'}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \in i\mathbb{R}^+, \end{aligned}$$

and

$$\begin{aligned}
 \mathcal{Y}_{b_1}^{[0]} &= 2 \int_{-\lambda_2}^{-\lambda_3} \frac{d\lambda}{y(\lambda)} + 2 \int_{\lambda_1}^{-\lambda_1} \frac{d\lambda}{y(\lambda)} = -\frac{2K'_1}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} + \frac{2(-iK'_1 - iK'_1)}{-i\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \\
 (7.64) \qquad \qquad \qquad &= \frac{2K'_1}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \in i\mathbb{R}^-.
 \end{aligned}$$

**Computation of Type 2 Integrals  $(\mathcal{Y}_{a_i}^{[1]}, \mathcal{Y}_{b_i}^{[1]})$ .**

Let  $\beta_1 < \beta_2 < \beta_3$  and introduce the substitution

$$(7.65) \qquad z^2 = \frac{\beta_1 - \beta_2}{\Lambda - \beta_3}, \quad \Lambda = \beta_3 + \frac{\beta_1 - \beta_3}{z^2}, \quad \Lambda = \lambda^2.$$

Differentiation yields

$$(7.66) \qquad d\Lambda = \frac{2(\beta_3 - \beta_1)}{z^3} dz.$$

Using (7.65) one checks

$$(7.67) \qquad \Lambda - \beta_1 = (\beta_3 - \beta_1) \frac{z^2 - 1}{z^2},$$

$$(7.68) \qquad \Lambda - \beta_2 = \frac{(\beta_3 - \beta_2)z^2 + (\beta_1 - \beta_3)}{z^2},$$

$$(7.69) \qquad \Lambda - \beta_3 = \frac{\beta_1 - \beta_3}{z^2}.$$

Consequently the cubic product factors as

$$(7.70) \qquad \prod_{i=1}^3 (\Lambda - \beta_i) = (\beta_3 - \beta_1)^3 \frac{(z^2 - 1)(1 - k_2^2 z^2)}{z^6},$$

with

$$(7.71) \qquad k_2^2 = \frac{\beta_3 - \beta_2}{\beta_3 - \beta_1} \in (0, 1).$$

Taking principal square roots gives

$$(7.72) \qquad \left( \prod_{i=1}^3 (\Lambda - \beta_i) \right)^{1/2} = (\beta_3 - \beta_1) \sqrt{\beta_1 - \beta_3} \frac{\sqrt{(1 - z^2)(1 - k_2^2 z^2)}}{z^3}.$$

Combining (7.66) and (7.72) we obtain the reduction

$$(7.73) \qquad \int \frac{d\Lambda}{\sqrt{\prod_{i=1}^3 (\Lambda - \beta_i)}} = \frac{2}{\sqrt{\beta_1 - \beta_3}} \int \frac{dz}{\sqrt{(1 - z^2)(1 - k_2^2 z^2)}}.$$

Specialize  $\beta_1 = \lambda_3^2$ ,  $\beta_2 = \lambda_2^2$ ,  $\beta_3 = \lambda_1^2$ . Then

$$(7.74) \qquad z^2 = \frac{\lambda_3^2 - \lambda_2^2}{\lambda^2 - \lambda_1^2}, \quad z = \frac{\sqrt{\lambda_1^2 - \lambda_3^2}}{\sqrt{\lambda_1^2 - \lambda^2}},$$

and

$$(7.75) \qquad k_2^2 = \frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2} \in (0, 1).$$

Choose the cut  $[-\lambda_1, \lambda_1]$  for  $\sqrt{\lambda_1^2 - \lambda^2}$  and fix the sheet by the condition

$$\sqrt{\lambda_1^2 - \lambda^2} \sim -i\lambda \quad (\lambda \rightarrow \infty).$$

For  $\lambda = it$  ( $t \in \mathbb{R}$ ) and  $\lambda_j = it_j$  one then has the pointwise values (for  $\text{Re } \lambda \geq 0$ )

$$(7.76) \quad \sqrt{\lambda_1^2 - \lambda^2} = \begin{cases} \sqrt{t^2 - t_1^2}, & t > t_1, \\ -i\sqrt{t_1^2 - t^2}, & -t_1 < t < t_1, \\ -\sqrt{t^2 - t_1^2}, & t < -t_1, \end{cases}$$

which are consistent with the chosen asymptotics.

Consequently, for  $\lambda \in i\mathbb{R}$  with  $\text{Re } \lambda \geq 0$ , the substitution takes values as follows

$$(7.77) \quad z = \frac{\sqrt{\lambda_1^2 - \lambda^2}}{\sqrt{\lambda_1^2 - \lambda_3^2}} \in \begin{cases} \mathbb{R}^+, & \lambda \in (\lambda_1, +i\infty), \\ i\mathbb{R}^+, & \lambda \in (-\lambda_1, \lambda_1), \\ \mathbb{R}^-, & \lambda \in (-i\infty, -\lambda_1). \end{cases}$$

The oriented segments in the  $\lambda$ -plane map to the  $z$ -plane under (7.77) as follows (principal images):

$$(7.78) \quad (+i\infty, \lambda_3) \longrightarrow (0, 1), \quad (\lambda_3, \lambda_2) \longrightarrow (1, k_2^{-1}), \quad (\lambda_2, \lambda_1) \longrightarrow (k_2^{-1}, +\infty),$$

$$(7.79) \quad (\lambda_1, 0) \longrightarrow \left(+i\infty, \frac{\sqrt{\lambda_1^2 - \lambda_3^2}}{-\lambda_1}\right), \quad (0, -\lambda_1) \longrightarrow \left(\frac{\sqrt{\lambda_1^2 - \lambda_3^2}}{-\lambda_1}, +\infty\right),$$

$$(7.80) \quad (-\lambda_1, -\lambda_2) \longrightarrow (-\infty, -k_2^{-1}), \quad (-\lambda_2, -\lambda_3) \longrightarrow (-k_2^{-1}, -1), \quad (-\lambda_3, -i\infty) \longrightarrow (-1, 0).$$

By the Type 2 reduction (with  $\Lambda = \lambda^2$ ) the cubic factor admits the representation

$$y(\lambda) = \prod_{i=1}^3 (\Lambda - \beta_i)^{1/2} = (\beta_3 - \beta_1) \sqrt{\beta_1 - \beta_3} \frac{\sqrt{(1 - z^2)(1 - k_2^2 z^2)}}{z^3}, \quad \beta_j = \lambda_{4-j}^2,$$

On the chosen sheet for  $\lambda \in i\mathbb{R}$  along the imaginary axis, comparison of signs yields

$$y(\lambda) = -i(\beta_3 - \beta_1) \sqrt{\beta_3 - \beta_1} \frac{f(z, k_2)}{z^3}, \quad f(z, k_2) = \sqrt{(1 - z^2)(1 - k_2^2 z^2)}.$$

where the branch cuts of  $f(z, k_2)$  are chosen as in (7.28)–(7.30). Hence, for  $\lambda \in i\mathbb{R}$ ,

$$\frac{\lambda d\lambda}{y(\lambda)} = \frac{1}{-i\sqrt{\beta_3 - \beta_1}} \frac{dz}{f(z, k_2)} = \frac{i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \frac{dz}{f(z, k_2)},$$

Introducing the complete elliptic integrals  $K_2 = K(k_2)$  and  $K'_2 = K(k'_2)$ , and using the standard substitutions along the indicated images, the period integrals (type 2 moments with a factor  $\lambda$ ) evaluate as follows.

$$(7.81) \quad \mathcal{Y}_{a_1}^{[1]} = 2 \int_{\lambda_2}^{\lambda_1} \frac{\lambda d\lambda}{y(\lambda)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{k_2^{-1}}^{+\infty} \frac{dz}{f(z, k_2)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K_2 + iK'_2}^{iK'_2} dv = \frac{-2i K_2}{\sqrt{\lambda_1^2 - \lambda_3^2}} \in i\mathbb{R}^-,$$

$$(7.82) \quad \mathcal{Y}_{a_2}^{[1]} = 2 \int_{-\lambda_1}^{-\lambda_2} \frac{\lambda d\lambda}{y(\lambda)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{-\infty}^{-k_2^{-1}} \frac{dz}{f(z, k_2)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{iK'_2}^{-K_2 + iK'_2} dv = \frac{-2i K_2}{\sqrt{\lambda_1^2 - \lambda_3^2}} \in i\mathbb{R}^-,$$

$$(7.83) \quad \mathcal{Y}_{b_2}^{[1]} = 2 \int_{-\lambda_2}^{-\lambda_3} \frac{\lambda d\lambda}{y(\lambda)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{-k_2^{-1}}^{-1} \frac{dz}{f(z, k_2)} = \frac{2i}{\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{-K_2 + iK'_2}^{-K_2} dv = \frac{2 K'_2}{\sqrt{\lambda_1^2 - \lambda_3^2}} \in \mathbb{R}^+,$$

$$(7.84) \quad \mathcal{Y}_{b_1}^{[1]} = 2 \int_{-\lambda_2}^{-\lambda_3} \frac{\lambda d\lambda}{y(\lambda)} + 2 \int_{\lambda_1}^{-\lambda_1} \frac{\lambda d\lambda}{y(\lambda)} = \mathcal{Y}_{b_2}^{[1]} = \frac{2 K'_2}{\sqrt{\lambda_1^2 - \lambda_3^2}} \in \mathbb{R}^+.$$

## Week 8: Algebraic-Geometric Solutions for the mKdV Equation (8)

2025-11-11

**Objectives:** Focus on Case 1. Compute  $\mathcal{Y}_{b_i}^{[0]}$ , and discuss how to determine the value of a multivalued complex function with a prescribed branch cut, taking  $y(\lambda) = \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}$  with the asymptotic condition  $y(\lambda) \sim \lambda^3$  as  $\lambda \rightarrow \infty$ .

We have

$$(8.1) \quad B = \begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} \begin{pmatrix} \oint_{b_1} \frac{d\lambda}{y} & \oint_{b_2} \frac{d\lambda}{y} \\ \oint_{b_1} \frac{\lambda d\lambda}{y} & \oint_{b_2} \frac{\lambda d\lambda}{y} \end{pmatrix} = \begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} \begin{pmatrix} \mathcal{Y}_{b_1}^{[0]} & 0 \\ 0 & \mathcal{Y}_{b_1}^{[1]} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

We can obtain

$$(8.2) \quad \int_0^\lambda \frac{\chi d\chi}{\sqrt{\prod_{i=1}^3 (\chi^2 - \lambda_i^2)}} = \frac{1}{\sqrt{\lambda_3^2 - \lambda_1^2}} \int_0^z \frac{dt}{\sqrt{(1-t^2)(1-k_2^{(1)2}t^2)}} = \frac{1}{\sqrt{\lambda_3^2 - \lambda_1^2}} \int_0^v dv.$$

and

$$(8.3) \quad \lambda^2 = \frac{\lambda_3^2 - \lambda_1^2 \operatorname{cn}^2(v, k_2^{(1)})}{\operatorname{sn}^2(v, k_2^{(1)})}, \quad \operatorname{sn}(v, k_2^{(1)}) = z = \sqrt{\frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda^2}}, \quad k_2^{(1)} = \sqrt{\frac{\lambda_1^2 - \lambda_2^2}{\lambda_1^2 - \lambda_3^2}}$$

and

$$v = F(z, k_2^{(1)}) = F\left(\sqrt{\frac{\lambda_1^2 - \lambda_3^2}{\lambda_1^2 - \lambda^2}}, k_2^{(1)}\right)$$

### Supplementary computations:

Let  $P = (\lambda, y)$  be a point on the Riemann surface defined by

$$y(\lambda) = \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}.$$

The first sheet is fixed by the asymptotic condition

$$y(\lambda) \sim \lambda^3 + O(\lambda), \quad \lambda \rightarrow \infty.$$

Set

$$(8.4) \quad \lambda_j = i t_j, \quad 0 < t_1 < t_2 < t_3, \quad j = 1, 2, 3.$$

For the chosen cut we rewrite  $y(\lambda)$  for  $\lambda = it \in i\mathbb{R}$  (keeping the factorised form with the fractional square roots):

$$(8.5) \quad \begin{aligned} y(\lambda) &= (\lambda - \lambda_3) \sqrt{\frac{\lambda - \lambda_2}{\lambda - \lambda_3}} (\lambda - \lambda_1) \sqrt{\frac{\lambda + \lambda_1}{\lambda - \lambda_1}} (\lambda + \lambda_2) \sqrt{\frac{\lambda + \lambda_3}{\lambda + \lambda_2}} \\ &= -i(t - t_3)(t - t_1)(t + t_2) \sqrt{\frac{t - t_2}{t - t_3}} \sqrt{\frac{t + t_1}{t - t_1}} \sqrt{\frac{t + t_3}{t + t_2}}. \end{aligned}$$

Here the square root is taken as the principal branch:

$$(8.6) \quad \sqrt{z} = \sqrt{|z|} e^{\frac{1}{2}i \arg z}, \quad \arg z \in (-\pi, \pi].$$

To determine the permutation of real/imaginary values and the signs of  $y(it)$  as  $t$  varies along the imaginary axis, it is convenient to count

- the sign  $s = \text{sign}((t - t_3)(t - t_1)(t + t_2))$  of the real linear prefactor, and
- the number  $k \in \{0, 1, 2, 3\}$  of the three real ratios

$$\frac{t - t_2}{t - t_3}, \quad \frac{t + t_1}{t - t_1}, \quad \frac{t + t_3}{t + t_2}$$

that are negative (each negative ratio contributes a factor  $i$  from the principal square root).

With these conventions, from (8.5) one has

$$y(it) = (-i)^s (i)^k (\text{positive real}) = -s i^{k+1} (\text{positive real}),$$

so that  $y(it)$  is imaginary when  $k \equiv 0, 2 \pmod{2}$  and real when  $k \equiv 1, 3 \pmod{2}$ , with the sign determined by the combination above.

Evaluating  $s$  and  $k$  on the natural partition of the real axis by  $\pm t_1, \pm t_2, \pm t_3$  gives the following classification:

$$(8.7) \quad y(it) \in \begin{cases} i\mathbb{R}^-, & t > t_3, \\ \mathbb{R}^-, & t_2 < t < t_3, \\ i\mathbb{R}^+, & t_1 < t < t_2, \\ \mathbb{R}^+, & -t_1 < t < -t_2, \\ i\mathbb{R}^-, & -t_2 < t < -t_1, \\ \mathbb{R}^-, & -t_3 < t < -t_2, \\ i\mathbb{R}^+, & t < -t_3. \end{cases}$$

For clarity, a short justification for each interval is as follows (sign of linear prefactor  $s$  and number  $k$  of negative ratios are given in parentheses):

- $t > t_3$ :  $s = +1, k = 0 \Rightarrow y = -i \cdot (+1) \cdot 1 \cdot (\text{pos}) \in i\mathbb{R}^-$ .
- $t_2 < t < t_3$ :  $s = -1, k = 1 \Rightarrow y = (-1) \cdot (\text{pos}) \in \mathbb{R}^-$ .
- $t_1 < t < t_2$ :  $s = -1, k = 0 \Rightarrow y = +i \cdot (\text{pos}) \in i\mathbb{R}^+$ .
- $-t_1 < t < -t_2$ :  $s = +1, k = 1 \Rightarrow y = +(\text{pos}) \in \mathbb{R}^+$ .
- $-t_2 < t < -t_1$ :  $s = +1, k = 0 \Rightarrow y = -i \cdot (\text{pos}) \in i\mathbb{R}^-$ .
- $-t_3 < t < -t_2$ :  $s = -1, k = 1 \Rightarrow y = (-1) \cdot (\text{pos}) \in \mathbb{R}^-$ .
- $t < -t_3$ :  $s = -1, k = 0 \Rightarrow y = +i \cdot (\text{pos}) \in i\mathbb{R}^+$ .

All statements above are obtained without algebraically combining the fractional square-root factors; each square root is treated as the principal branch (8.6) applied to the linear fractional arguments appearing in (8.5).

Near each branch point  $\lambda_j$  or  $\lambda_j^* = -\lambda_j$  the function behaves locally as  $\sqrt{\lambda \mp \lambda_j} \cdot (\text{analytic part})$ , and starting from  $\lambda = -i\infty$  (where  $y \in i\mathbb{R}^+$ ), moving upward along the right side of the imaginary axis one acquires at each branch point an additional phase  $i\pi/2$ , equivalently a multiplicative factor  $i$ , fully consistent with the above sign computation.

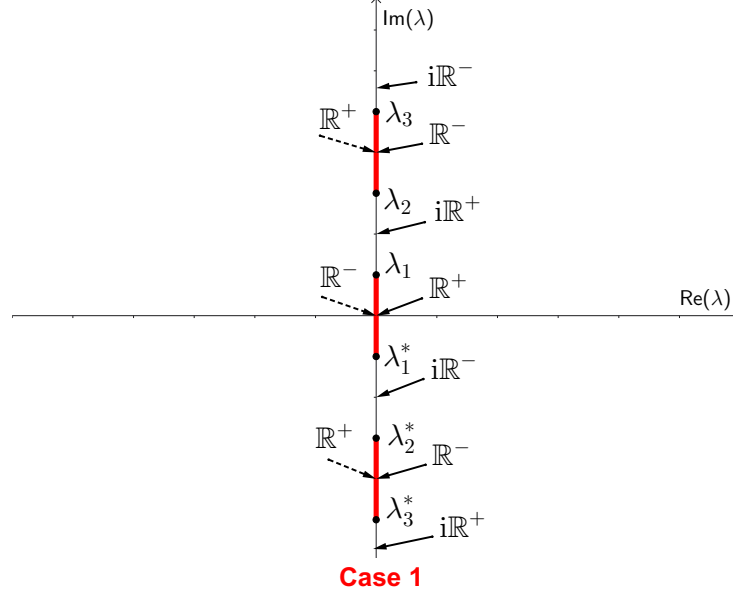


FIGURE 6. The sign values of  $y(\lambda) = \sqrt{(\lambda^2 - \lambda_1^2)(\lambda^2 - \lambda_2^2)(\lambda^2 - \lambda_3^2)}$ , with  $\lambda \rightarrow \infty$ ,  $y \rightarrow \lambda^3 + O(\lambda)$  on the first sheet for Case 1.

## Week 9: Algebraic-Geometric Solutions for the mKdV Equation (9)

2025-11-18

**Objectives:** Focus on Case 1. Introduce the Joukowski function to parametrize the two transformations used in the computation of  $\mathcal{Y}_{a_i}^{[j]}$ ; compute  $\mathcal{Y}_{b_i}^{[1]}$  and  $B_{ij}$ ; and discuss how to determine a suitable ansatz for the differentials  $d\Omega_j$  for  $j = 1, 2, 3$ .

### Supplementary computations:

The period normalization for the holomorphic differentials

$$w_i(\lambda) = \sum_{k=0}^1 d_{ik} \lambda^k y^{-1}$$

is expressed by

$$2\pi i \delta_{ij} = \oint_{a_j} w_i d\lambda = \sum_{k=0}^1 d_{ik} \mathcal{Y}_{a_j}^{[k]}, \quad B_{ij} = \oint_{b_j} w_i d\lambda = \sum_{k=0}^1 d_{ik} \mathcal{Y}_{b_j}^{[k]}.$$

The symmetry properties of the period integrals,

$$\mathcal{Y}_{a_1}^{[0]} = -\mathcal{Y}_{a_2}^{[0]}, \quad \mathcal{Y}_{b_1}^{[0]} = -\mathcal{Y}_{b_2}^{[0]}, \quad \mathcal{Y}_{a_1}^{[1]} = \mathcal{Y}_{a_2}^{[1]}, \quad \mathcal{Y}_{b_1}^{[1]} = \mathcal{Y}_{b_2}^{[1]},$$

allow the normalization conditions to be written as

$$(9.1) \quad \begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} \begin{pmatrix} \mathcal{Y}_{a_1}^{[0]} & 0 \\ 0 & \mathcal{Y}_{a_2}^{[1]} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = 2\pi i \mathbb{I}_2,$$

and

$$(9.2) \quad \begin{pmatrix} d_{10} & d_{11} \\ d_{20} & d_{21} \end{pmatrix} \begin{pmatrix} \mathcal{Y}_{b_1}^{[0]} & 0 \\ 0 & \mathcal{Y}_{b_2}^{[1]} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = B^{(1)}.$$

Eliminating the coefficients  $d_{ik}$  between (9.1) and (9.2) gives

$$(9.3) \quad \begin{aligned} B^{(1)} &= \pi i \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} (\mathcal{Y}_{a_1}^{[0]})^{-1} \mathcal{Y}_{b_1}^{[0]} & 0 \\ 0 & (\mathcal{Y}_{a_2}^{[1]})^{-1} \mathcal{Y}_{b_2}^{[1]} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \\ &= \pi i \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \tau_1^{(1)} & 0 \\ 0 & \tau_2^{(1)} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \\ &= \pi i \begin{pmatrix} \tau_1^{(1)} & \tau_2^{(1)} \\ -\tau_1^{(1)} & \tau_2^{(1)} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \pi i \begin{pmatrix} \tau_1^{(1)} + \tau_2^{(1)} & \tau_2^{(1)} - \tau_1^{(1)} \\ \tau_2^{(1)} - \tau_1^{(1)} & \tau_1^{(1)} + \tau_2^{(1)} \end{pmatrix}. \end{aligned}$$

Here the reduced moduli are given by

$$(9.4) \quad \tau_1^{(1)} = (\mathcal{Y}_{a_1}^{[0]})^{-1} \mathcal{Y}_{b_1}^{[0]} = i \frac{K_1^{(1)'}}{K_1^{(1)}}, \quad \tau_2^{(1)} = (\mathcal{Y}_{a_2}^{[1]})^{-1} \mathcal{Y}_{b_2}^{[1]} = i \frac{K_2^{(1)'}}{K_2^{(1)}}.$$

By the definition of the Riemann theta function,

$$(9.5) \quad \begin{aligned} \Theta(B^{(1)}; z) &= \sum_{n \in \mathbb{Z}^2} \exp \left\{ \frac{1}{2} n^\top B^{(1)} n + n^\top z \right\} \\ &= \sum_{n \in \mathbb{Z}^2} \exp \left\{ \frac{\pi i}{2} n^\top \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \tau_1^{(1)} & 0 \\ 0 & \tau_2^{(1)} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} n + n^\top z \right\}, \end{aligned}$$

where

$$(9.6) \quad n = \begin{pmatrix} n_1 \\ n_2 \end{pmatrix}, \quad z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}, \quad n_1, n_2 \in \mathbb{Z}, \quad z_1, z_2 \in \mathbb{C}.$$

Decompose  $n$  as

$$(9.7) \quad n = \begin{pmatrix} n_1 \\ n_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \end{pmatrix} + \begin{pmatrix} t \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} m_1 + \frac{t}{2} \\ m_2 + \frac{t}{2} \end{pmatrix}, \quad t = 0, 1,$$

so that

$$(9.8) \quad n_1 = m_1 + m_2 + t = 2m_1 + n_2 + t, \quad n_2 = m_2 - m_1, \quad m_1, m_2 \in \mathbb{Z}.$$

As  $(m_1, m_2)$  range over  $\mathbb{Z}^2$ , so do  $(n_1, n_2)$ .

Substituting (9.7) into (9.5) yields

$$(9.9) \quad \begin{aligned} \Theta(B^{(1)}; z) &= \sum_{t=0,1} \sum_{m_1, m_2 \in \mathbb{Z}} \exp \left\{ 2\pi i \left( m_1 + \frac{t}{2}, m_2 + \frac{t}{2} \right) \begin{pmatrix} \tau_1^{(1)} & 0 \\ 0 & \tau_2^{(1)} \end{pmatrix} \begin{pmatrix} m_1 + \frac{t}{2} \\ m_2 + \frac{t}{2} \end{pmatrix} \right. \\ &\quad \left. + \left( m_1 + \frac{t}{2}, m_2 + \frac{t}{2} \right) \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \right\} \\ &= \sum_{t=0,1} \sum_{m_1, m_2 \in \mathbb{Z}} \exp \left\{ 2\pi i \left( m_1 + \frac{t}{2} \right)^2 \tau_1^{(1)} + 2\pi i \left( m_2 + \frac{t}{2} \right)^2 \tau_2^{(1)} \right. \\ &\quad \left. + \left( m_1 + \frac{t}{2} \right) (z_1 - z_2) + \left( m_2 + \frac{t}{2} \right) (z_1 + z_2) \right\} \\ &= \vartheta_3(z_1 - z_2; 2\tau_1^{(1)}) \vartheta_3(z_1 + z_2; 2\tau_2^{(1)}) + \vartheta_2(z_1 - z_2; 2\tau_1^{(1)}) \vartheta_2(z_1 + z_2; 2\tau_2^{(1)}). \end{aligned}$$



**Definition 3.** The Jacobi theta functions are defined by

$$(9.10) \quad \begin{aligned} \vartheta_1(u, \tau) &:= \Theta \begin{bmatrix} 1 \\ 1 \end{bmatrix} (u, \tau), & \vartheta_2(u, \tau) &:= \Theta \begin{bmatrix} 1 \\ 0 \end{bmatrix} (u, \tau), \\ \vartheta_3(u, \tau) &:= \Theta \begin{bmatrix} 0 \\ 0 \end{bmatrix} (u, \tau), & \vartheta_4(u, \tau) &:= \Theta \begin{bmatrix} 0 \\ 1 \end{bmatrix} (u, \tau), \end{aligned}$$

with  $\text{Im } \tau > 0$  and

$$(9.11) \quad \Theta \begin{bmatrix} \epsilon \\ \epsilon' \end{bmatrix} (u, \tau) := \sum_{n \in \mathbb{Z}} \exp \left\{ \pi i \left( n + \frac{\epsilon}{2} \right)^2 \tau + \left( n + \frac{\epsilon}{2} \right) (u + \pi i \epsilon') \right\}, \quad \epsilon, \epsilon' \in \{0, 1\}.$$

## Week 10: Algebro-Geometric Solutions for the mKdV Equation (10)

2025-11-25

**Objectives:** Focus on Case 1. Compute  $c_{2i}, c_{3,i}$  and  $\Omega_j^{(1)}(P)$ .

**Supplementary computations:**

**Computation of Type 1 Integrals  $(\mathcal{Y}_{a_i}^{[2]}, \mathcal{Y}_{b_i}^{[2]})$ .**

We introduce the auxiliary function  $\Lambda$  (cf. (7.42)) by

$$(10.1) \quad \Lambda(z) = \frac{\lambda_1^2 z^2}{z^2 - \frac{\lambda_1^2 - \lambda_3^2}{-\lambda_3^2}} = \frac{\lambda_1^2 \lambda_3^2 z^2}{\lambda_3^2 z^2 + (\lambda_1^2 - \lambda_3^2)} = \frac{\lambda_1^2 \lambda_3^2 z^2}{\lambda_1^2 - \lambda_3^2 (1 - z^2)}.$$

Recall the relation between  $z$  and the elliptic argument  $\nu$ ,

$$(10.2) \quad \nu = F(z, k_1^{(1)}), \quad z = \text{sn}(\nu, k_1^{(1)}).$$

We will use the imaginary-period shift identities for Jacobi elliptic functions,

$$(10.3) \quad \text{sn}(\nu + iK_1', k_1^{(1)}) = \frac{1}{k_1^{(1)} \text{sn}(\nu, k_1^{(1)})}, \quad \text{cn}(\nu + iK_1', k_1^{(1)}) = -i \frac{\text{dn}(\nu, k_1^{(1)})}{k_1^{(1)} \text{sn}(\nu, k_1^{(1)})}.$$

Set

$$(10.4) \quad \nu = \nu_1 + iK_1',$$

so that  $\text{sn}(\nu, k_1^{(1)})$  and  $\text{cn}(\nu, k_1^{(1)})$  in (10.1) may be expressed in terms of  $\nu_1$ . Substituting  $z = \text{sn}(\nu, k_1^{(1)})$  into (10.1) and using (10.3) yields

$$(10.5) \quad \begin{aligned} \Lambda &= \frac{\lambda_1^2 \lambda_3^2 \text{sn}^2(\nu, k_1^{(1)})}{\lambda_1^2 - \lambda_3^2 \text{cn}^2(\nu, k_1^{(1)})} = \frac{\lambda_1^2 \lambda_3^2}{\lambda_1^2 (k_1^{(1)})^2 \text{sn}^2(\nu_1, k_1^{(1)}) + \lambda_3^2 \text{dn}^2(\nu_1, k_1^{(1)})} \\ &= \frac{\lambda_1^2 \lambda_3^2}{\lambda_3^2 + (k_1^{(1)})^2 (\lambda_1^2 - \lambda_3^2) \text{sn}^2(\nu_1, k_1^{(1)})}. \end{aligned}$$

Using the identity

$$(k_1^{(1)})^2 = \frac{(\lambda_1^2 - \lambda_2^2) \lambda_3^2}{(\lambda_1^2 - \lambda_3^2) \lambda_2^2},$$

the expression (10.5) can be rewritten in the equivalent forms

$$(10.6) \quad \Lambda = \frac{\lambda_1^2 \lambda_3^2}{\lambda_3^2 + \frac{\lambda_3^2}{\lambda_2^2} (\lambda_1^2 - \lambda_2^2) \text{sn}^2(\nu_1, k_1^{(1)})} = \frac{\lambda_1^2}{1 - \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2} \text{sn}^2(\nu_1, k_1^{(1)})}.$$

The last form highlights that  $\Lambda$  becomes a simple rational function of  $\text{sn}^2(\nu_1, k_1^{(1)})$ , which is convenient for integrating.

We now compute  $\mathcal{Y}_{a_1}^{[2]}$ . Starting from the integral in (type 1) and changing variables to the elliptic parameter  $\nu$  (with the shift  $\nu = \nu_1 + iK'_1$ ) one obtains

$$(10.7) \quad \mathcal{Y}_{a_1}^{[2]} = 2 \int_{\lambda_2}^{\lambda_1} \frac{\lambda^2 d\lambda}{y(\lambda)} = \frac{2\lambda_1^2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_{K_1 + iK'_1}^{iK'_1} \frac{d\nu}{1 - \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2} \operatorname{sn}^2(\nu - iK'_1, k_1^{(1)})}.$$

The integrand is an elementary function of  $\operatorname{sn}^2$ , hence the integral is an incomplete elliptic integral of the third kind. Writing the integral in terms of  $\Pi(\alpha^2, u, k)$ , with characteristic  $\alpha_1^2 = \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2}$ , modulus  $k = k_1^{(1)}$ , and upper limit  $u$ , we have

$$(10.8) \quad \mathcal{Y}_{a_1}^{[2]} = \frac{2\lambda_1^2}{-i\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \Pi\left(\alpha_1^2, u, k_1^{(1)}\right) \Big|_{u=K_1}^{u=0}, \quad \alpha_1^2 = \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2}.$$

Evaluating the difference of the incomplete elliptic integrals between  $u = K_1$  and  $u = 0$  gives the complete elliptic integral of the third kind  $\Pi(\alpha_1^2, k)$ . Therefore

$$(10.9) \quad \mathcal{Y}_{a_1}^{[2]} = \frac{-2i\lambda_1^2}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \Pi(\alpha_1^2, k_1^{(1)}), \quad \alpha_1^2 = \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2}.$$

By symmetry of the contour (or by the corresponding sign change in the integration limits), one obtains the companion relation

$$(10.10) \quad \mathcal{Y}_{a_2}^{[2]} = -\mathcal{Y}_{a_1}^{[2]}.$$

Similarly,

$$(10.11) \quad \begin{aligned} \mathcal{Y}_{b_2}^{[2]} &= 2 \int_{-\lambda_2}^{-\lambda_3} \frac{\lambda^2 d\lambda}{y(\lambda)} = \frac{2\lambda_1^2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_{K_1}^{K_1 + iK'_1} \frac{d\nu}{1 - \alpha_1^2 \operatorname{sn}^2(\nu - iK'_1, k_1^{(1)})} \\ &= \frac{2\lambda_1^2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \Pi(\alpha_1^2, u, k_1^{(1)}) \Big|_{u=K_1 - iK'_1}^{u=K_1}, \quad \alpha_1^2 = \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2}. \end{aligned}$$

We use the standard Jacobi identities

$$(10.12) \quad \operatorname{sn}(z + K, k) = \frac{\operatorname{cn}(z, k)}{\operatorname{dn}(z, k)}, \quad \operatorname{sn}(iz, k) = i \frac{\operatorname{sn}(z, k')}{\operatorname{cn}(z, k')},$$

$$(10.13) \quad \operatorname{cn}(iz, k) = \frac{1}{\operatorname{cn}(z, k')}, \quad \operatorname{dn}(iz, k) = \frac{\operatorname{dn}(z, k')}{\operatorname{cn}(z, k')},$$

which imply

$$(10.14) \quad \operatorname{dn}(z - K'_1, k'_1) = \frac{k_1}{\operatorname{dn}(z, k'_1)}, \quad \operatorname{sn}(iz + K_1, k) = \frac{1}{\operatorname{dn}(z, k')}.$$

Hence,

$$(10.15) \quad \operatorname{sn}(iu + K_1 - iK'_1, k_1^{(1)}) = \frac{1}{\operatorname{dn}(u - K'_1, k_1^{(1)'})} = \frac{\operatorname{dn}(u, k_1^{(1)'})}{k_1^{(1)}}.$$

Let  $v = iu + K_1$ . Then

$$\begin{aligned}
\mathcal{Y}_{b_2}^{[2]} &= \frac{2\lambda_1^2}{\sqrt{-\lambda_2^2(\lambda_1^2 - \lambda_3^2)}} \int_0^{K'_1} \frac{i du}{1 - \alpha_1^2 \operatorname{sn}^2(iu + K_1 - iK'_1, k_1^{(1)})} \\
(10.16) \quad &= \frac{2\lambda_1^2}{-i\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \int_0^{K'_1} \frac{i du}{1 - \alpha_1^2 \left( \frac{1}{(k_1^{(1)})^2} - \left( \frac{k_1^{(1)'}}{k_1^{(1)}} \right)^2 \operatorname{sn}^2(u, k_1^{(1)'}) \right)} \\
&= \frac{-2\lambda_3^2}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \int_0^{K'_1} \frac{du}{1 - \alpha_2^2 \operatorname{sn}^2(u, k_1^{(1)'})} = \frac{-2\lambda_3^2}{\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \Pi(\alpha_2^2, k_1^{(1)'}), \quad \alpha_2^2 = \frac{\lambda_2^2 - \lambda_3^2}{\lambda_2^2}.
\end{aligned}$$

One has

$$\begin{aligned}
\mathcal{Y}_{b_1}^{[2]} &= 2 \int_{-\lambda_2}^{-\lambda_3} \frac{\lambda^2 d\lambda}{y(\lambda)} + 2 \int_{\lambda_1}^{-\lambda_1} \frac{\lambda^2 d\lambda}{y(\lambda)} \\
(10.17) \quad &= \mathcal{Y}_{b_2}^{[2]} + \frac{2\lambda_1^2}{-i\lambda_2 \sqrt{\lambda_1^2 - \lambda_3^2}} \int_{iK'_1}^{-iK'_1} \frac{dv}{1 - \alpha_1^2 \operatorname{sn}^2(v - iK'_1, k_1^{(1)})}.
\end{aligned}$$

We simplify the elliptic integrand using standard shifts of Jacobi functions. For complementary modulus  $k'$  one has

$$\operatorname{sn}(u - K', k') = -\frac{\operatorname{cn}(u, k')}{\operatorname{dn}(u, k')}, \quad \operatorname{cn}(u - K', k') = \frac{k \operatorname{sn}(u, k')}{\operatorname{dn}(u, k')}.$$

From these identities it follows that,

$$\operatorname{sn}(iu - iK'_1, k_1^{(1)}) = i \frac{\operatorname{sn}(u - K'_1, k_1^{(1)'})}{\operatorname{cn}(u - K'_1, k_1^{(1)'})} = -\frac{i}{k_1^{(1)}} \frac{\operatorname{cn}(u, k_1^{(1)'})}{\operatorname{sn}(u, k_1^{(1)'})}.$$

Now set  $v = iu$  and substitute the last expression into the integrand. Using

$$\alpha_1^2 = \frac{\lambda_1^2 - \lambda_2^2}{-\lambda_2^2}, \quad (k_1^{(1)})^2 = \frac{(\lambda_1^2 - \lambda_2^2)\lambda_3^2}{(\lambda_1^2 - \lambda_3^2)\lambda_2^2},$$

we compute the denominator appearing in the integrand

$$\begin{aligned}
1 - \alpha_1^2 \operatorname{sn}^2(v - iK'_1, k_1^{(1)}) &= 1 - \frac{\lambda_1^2 - \lambda_2^2}{\lambda_2^2} \frac{1}{(k_1^{(1)})^2} \frac{\operatorname{cn}^2(u, k_1^{(1)'})}{\operatorname{sn}^2(u, k_1^{(1)'})} = 1 - \frac{\lambda_1^2 - \lambda_3^2}{\lambda_3^2} \frac{\operatorname{cn}^2(u, k_1^{(1)'})}{\operatorname{sn}^2(u, k_1^{(1)'})} \\
(10.18) \quad &= \frac{\lambda_3^2 \operatorname{sn}^2(u, k_1^{(1)'}) - (\lambda_1^2 - \lambda_3^2) \operatorname{cn}^2(u, k_1^{(1)'})}{\lambda_3^2 \operatorname{sn}^2(u, k_1^{(1)'})} = \frac{\lambda_1^2 \operatorname{sn}^2(u, k_1^{(1)'}) - (\lambda_1^2 - \lambda_3^2)}{\lambda_3^2 \operatorname{sn}^2(u, k_1^{(1)'})}.
\end{aligned}$$

This expression for the denominator is used to rewrite the integral in (10.17) in terms of elliptic functions with argument  $u$ . From (10.18) we obtain the partial-fraction-type decomposition

$$\begin{aligned}
\frac{1}{1 - \alpha_1^2 \operatorname{sn}^2(v - iK'_1, k_1^{(1)})} &= \frac{\lambda_3^2}{\lambda_1^2} + \frac{\frac{\lambda_3^2}{\lambda_1^2} (\lambda_1^2 - \lambda_3^2)}{\lambda_1^2 \operatorname{sn}^2(u, k_1^{(1)'}) - (\lambda_1^2 - \lambda_3^2)} \\
(10.19) \quad &= \frac{\lambda_3^2}{\lambda_1^2} - \frac{\lambda_3^2}{\lambda_1^2} \frac{1}{1 - \frac{\lambda_1^2}{\lambda_1^2 - \lambda_3^2} \operatorname{sn}^2(u, k_1^{(1)'})}.
\end{aligned}$$

Hence the relation between  $\mathcal{Y}_{b_1}^{[2]}$  and  $\mathcal{Y}_{b_2}^{[2]}$  follows:

$$\begin{aligned}
 \mathcal{Y}_{b_1}^{[2]} &= \mathcal{Y}_{b_2}^{[2]} + \frac{2\lambda_3^2}{-i\lambda_2\sqrt{\lambda_1^2 - \lambda_3^2}} \int_{K_1'}^{-K_1'} \left( 1 - \frac{1}{1 - \frac{\lambda_1^2}{\lambda_1^2 - \lambda_3^2} \operatorname{sn}^2(u, k_1^{(1)'})} \right) i du \\
 (10.20) \quad &= \mathcal{Y}_{b_2}^{[2]} + \frac{4\lambda_3^2 K_1'}{\lambda_2\sqrt{\lambda_1^2 - \lambda_3^2}} - \frac{4\lambda_3^2}{\lambda_2\sqrt{\lambda_1^2 - \lambda_3^2}} \Pi\left(\frac{\lambda_1^2}{\lambda_1^2 - \lambda_3^2}, k_1^{(1)'}\right) \\
 &= \mathcal{Y}_{b_2}^{[2]} + 2\lambda_3^2 \mathcal{Y}_{b_1}^{[0]} - \frac{4\lambda_3^2}{\lambda_2\sqrt{\lambda_1^2 - \lambda_3^2}} \Pi\left(a_3^2, k_1^{(1)'}\right), \quad a_3^2 = \frac{\lambda_1^2}{\lambda_1^2 - \lambda_3^2},
 \end{aligned}$$

where the middle line uses the definition of the complete elliptic integral of the third kind and the last line expresses the  $K_1'$ -term via  $\mathcal{Y}_{b_1}^{[0]}$ .

## Week 11: Algebro-Geometric Solutions for the mKdV Equation (11)

2025-12-02

**Objectives:** Focus on Case 1. Determine the parameters  $C_{u_0}^{(1)}$  and  $D^{(1)}$ .

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